

What Held-Out Accuracy Predicts About Deploying Intrusion Detection in O-RAN

ARSHAD ALI¹ AND ADEEL AHMAD¹

¹Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia

ABSTRACT Intrusion detectors for 5G and O-RAN traffic are frequently reported to exceed 99% accuracy on a random split of a single corpus. This paper examines what such results predict about deployment, using NetsLab-5GORAN-IDD, 5G-NIDD, six architectures, two input-blind baselines, and a ladder of increasingly strict protocols. A random split raises radio-layer macro- F_1 by 0.13 (95% interval -0.01 to 0.27; significant for two sequence models), and a nearest-neighbor lookup gains 0.15, consistent with the recognition of capture sessions. Because the radio capture is ordered by scenario, a time-disjoint split excludes entire attack categories from training. When category coverage is held fixed, temporal order cannot be distinguished from session-to-session variation, and the false positive rate on a held-out benign session ranges from 0.00 to 0.98. In 5G-NIDD, 59% of the benign flows duplicate flood records labeled as attacks elsewhere, limiting balanced accuracy to 0.766. The 99.9% accuracy reported by the dataset authors is reproducible but depends on two record-position fields, without which accuracy decreases to 76.9%. On 5G-NIDD flows with consistent labels, transferred detectors achieve a balanced accuracy of 0.61 to 0.78, below the in-target reference, and none satisfies a deployability test that the published pipeline satisfies on its own split. A third corpus exhibits the same behavior. At a prevalence of 0.002, operational precision does not exceed 0.063 at any threshold that retains recall above 10%. Averaging precision across folds inflates the best-to-worst ratio between architectures from 1.45 to 16.0. The paper concludes with reporting requirements for deployability claims.

INDEX TERMS Dataset shift, evaluation methodology, false positive rate, intrusion detection, label quality, O-RAN, RAN intelligent controller, xApp.

I. INTRODUCTION

Open Radio Access Network (O-RAN) architectures dis-aggregate the base station into radio, distributed, and central units that are connected through open interfaces, and they introduce a RAN Intelligent Controller (RIC) that hosts third-party control applications [1]–[3]. Because the near-real-time RIC operates close to the cells it controls [1], [4], it has access to the telemetry of every device served by those cells. Compromised devices, many of which are IoT devices, are a well-documented source of volumetric attacks [5], [6]. A detection application (xApp) placed in the RIC control loop is therefore an attractive defensive option. Platforms for developing xApps are available [7], [8], and detectors that operate within or alongside the O-RAN control loop have already been demonstrated [9], [10].

Detectors for 5G and O-RAN traffic are typically evaluated on a random held-out split of a single dataset, and the reported accuracy or F_1 score commonly exceeds 99%. For 5G-NIDD, one of the two corpora used in this study, the dataset authors report binary accuracies between 99.85% and 99.95% for four of five classifiers [11], and subsequent work

reports a weighted F_1 score of up to 99.95% [12]. However, whether such a detector can be deployed depends on three properties that a held-out split of the training distribution does not measure. The first is generalization to a deployment that did not contribute training data, and intrusion detectors are known to transfer poorly between independently collected datasets [13], [14]. The second is the alert burden, since a false positive rate that appears small on a balanced benchmark produces a large volume of alerts once it is multiplied by production flow rates and interpreted against a realistic attack base rate [15], [16]. The third is latency relative to the near-real-time control loop, for which the O-RAN E2 specification requires support for latencies between 10 ms and 1 s [17]. A benchmark score thus indicates how well a detector separates the records of one corpus, whereas deployment concerns its behavior on traffic it was not trained on, at a realistic base rate, and within a control loop. In this paper, benchmark results are treated as evidence about deployment only when an experiment supports that interpretation.

These concerns have been raised in earlier work. The base-rate argument was made by Axelsson [15] and the closed-

world critique by Sommer and Paxson [16]. Common pitfalls of machine learning in security were catalogued by Arp et al. [18], and temporal and spatial bias were formalized by Pendlebury et al. [19]. Transfer between the two corpora used here has been measured by Abraheem and Edhirig [20], and inference inside a real near-real-time RIC has been measured by Obiuevwi et al. [21]. The contribution of this paper is neither a new detector nor the observation that transfer degrades performance. It is instead a single controlled evaluation of one detector family in which these effects are measured jointly, with input-blind baselines evaluated on every corpus, so that their relative magnitudes can be compared and the claims that commonly rely on them can be tested. The specific contributions are as follows.

- We evaluate a ladder of evaluation protocols on a single pipeline with six architectures and two input-blind baselines. A model-free nearest-neighbor probe and two sequence models indicate that the random-split gain arises from the recognition of capture sessions (Section VI-A). A split that fixes category coverage further shows that the radio corpus does not support an estimate of temporal drift and that the false positive rate depends mainly on which benign session is tested (Section VI-B).
- We audit the target corpus and show that half of the 5G-NIDD flows lie on records that occur under both labels and that a published 99.9% result on this corpus depends on record-position fields. We then evaluate the published pipeline under the same ladder of protocols (Section VI-C).
- We separate the contributions of prior shift, label conflicts, and transfer failure to the cross-corpus gap by means of a prevalence-invariant metric, a difference-in-differences test against the majority baseline, references measured within the target corpus, and evaluations with and without the conflicting records. The transfer experiment is also repeated on a third corpus collected at a different site with a different exporter (Sections VI-D, VI-E, and VI-F).
- We report operational precision and alert volume from pooled confusion counts, in explicit units and with cluster-bootstrap intervals, and we show that averaging precision over evaluation folds inflates the apparent difference between architectures by an order of magnitude (Section VI-G).
- We identify which calibration methods can alter the achievable operating points, show that none of them improves these operating points in the present setting, and show that label-free estimation of the target prior fails when the class-conditional distributions shift (Section VI-H).
- We decompose decision latency on one host for three implementations, including a compiled runtime (Section VI-I). Finally, we combine generalization, alert burden, and latency into a deployability test, examine how its verdict depends on its bounds, and derive reporting

TABLE 1. Notation and abbreviations.

Symbol	Meaning
$\mathcal{D}_A, \mathcal{D}_B$	source corpus (NetsLab-5GORAN-IDD); target corpus (5G-NIDD)
$\mathcal{D}_C$	third corpus, a second target (5G core datasets of [22])
τ	decision threshold on the attack score $\hat{p}$
$R(\tau), \text{FPR}(\tau)$	recall and false positive rate at τ
π	operational attack prevalence (declared, not measured)
π_{tr}	attack prevalence of the training data
λ_b	benign arrival rate (flows or windows per hour)
$A(\tau)$	false alerts per hour at τ
$\text{PPV}(\tau)$	operational precision at prevalence π
Δ_{BA}, Δ_{F1}	source minus target balanced accuracy, macro- F_1
$\bar{W}_1$	mean first-order Wasserstein distance over features
$L, q_p(L)$	decision latency and its p -th quantile
B, ε	latency budget and tolerated violation rate
$\delta, A_{\max}, r_{\min}, \rho$	bounds of the deployability test, Eq. (6)
δ_m	deadline margin in Algorithm 1 (design, not evaluated)
BA	balanced accuracy, the mean of TPR and TNR
LR, DT, RF	logistic regression, decision tree, random forest
XGB, HGB, MLP	XGBoost, histogram gradient boosting, multilayer perceptron
IF, AE, GRU	isolation forest, autoencoder, gated recurrent unit
KPM, RC	E2 service models for measurements and RAN control
UE	user equipment

requirements (Sections VII and X).

All latency measurements in this paper are emulated, and no RIC is present in the measurement path. In addition, the three corpora were produced by three different flow exporters, so each cross-corpus gap combines a change of deployment with a change of tooling; Section VI-E quantifies the effect of this confound for $\mathcal{D}_A$ and $\mathcal{D}_B$. The notation used throughout the paper is summarized in Table 1.

II. BACKGROUND AND RELATED WORK

This section reviews four bodies of work that are relevant to the present study: the RIC as a platform for detection, the security of O-RAN, datasets and learning methods for network intrusion detection, and the literature on evaluation practice and dataset shift on which our methodology draws.

A. THE NEAR-REAL-TIME RIC AS A DETECTION PLATFORM

xApps subscribe to E2 nodes through service models. Among these, E2SM-KPM delivers periodic key performance measurements, and E2SM-RC exposes control actions [1]. Platforms such as OpenRAN Gym [23], SCOPE [24], CoO-RAN [7], and FlexRIC [8], together with open radio stacks [25], [26], have made the development of experimental xApps practical. Because E2 carries aggregated measurements rather than packets, the data path of a flow-level detector requires separate consideration, which is discussed in Section VII-A.

B. SECURITY OF OPEN RAN

The threat model of the O-RAN Alliance [27] and several independent analyses [28], [29] identify the RIC and third-party xApps as a new attack surface, and security mechanisms across the O-RAN interfaces have been implemented and evaluated [30]. With respect to detection, 5G-Spector provides layer-3 attack detection as an O-RAN compliant

service [9], Det-RAN detects attacks from cross-layer data in real time [10], and Fard et al. examine the fusion of radio telemetry and flow records on the same source corpus that is used in this paper [31]. These studies propose or evaluate detectors within a single deployment. The present study asks, instead, what evidence would be needed to show that such a detector remains effective in a different deployment.

C. MACHINE LEARNING FOR NETWORK INTRUSION DETECTION

A number of IoT-oriented benchmark corpora are available [32]–[36]. Cellular corpora are less common. 5G-NIDD was captured on an operational 5G test network [37], [38], and NetsLab-5GORAN-IDD combines flow records with radio telemetry collected on an O-RAN testbed [39]. For 5G-NIDD, the dataset authors report binary accuracies of 99.85% to 99.95% for four of five classifiers on a random 70/30 split [11], and later work reports a weighted F_1 score of up to 99.95% [12]. Standardized NetFlow feature sets have made cross-corpus experiments feasible [40], [41]. The quality of benchmark corpora is, however, itself a source of error. Labeling and flow-construction faults have been documented in CIC-IDS2017 [42], and design flaws that allow a classifier to score well without learning the attack have been found repeatedly across NIDS benchmarks [43].

D. EVALUATION PRACTICE AND DATASET SHIFT

Intrusion detection results are often overly optimistic because of closed-world evaluation [16], neglect of the base rate [15], sampling bias and data leakage [18], [44], and temporal or spatial bias in the data split [19]. Apruzzese et al. examine the gap between reported and operational effectiveness across security domains [45], and Kus et al. observe a similar pattern in industrial intrusion detection when detectors are re-evaluated on data on which they were not tuned [46]. The cross-dataset generalization of such detectors has repeatedly been found to be poor [13], [14], [47]. The work most closely related to ours is that of Abraheem and Edhirig, who transfer XGBoost between the same two corpora using 15 harmonized features and report zero-shot balanced accuracies per attack family of 0.50 to 0.56 from NetsLab-5GORAN-IDD to 5G-NIDD and of 0.52 to 0.83 in the reverse direction [20]. Our transfer results are consistent with theirs in both direction and magnitude. Relative to that study, the present paper contributes the protocol ladder with input-blind baselines, reference results measured within the target corpus, the evaluation of a published pipeline under the same ladder, an audit of the target labels, and the operational and latency criteria. We also draw on methods from the dataset-shift literature, namely label-shift estimation from unlabeled data [48], [49], calibration and its behavior under shift [50]–[52], shift detection [53], and covariance alignment for domain adaptation [54]. For the comparison of classifiers under repeated resampling, we follow Nadeau and Bengio [55].

III. DEPLOYMENT CRITERIA AND THREAT MODEL

This section describes the system and threat model, defines the three quantities that the evaluation measures, and combines them into a deployability test.

A. SYSTEM AND THREAT MODEL

A near-real-time RIC serves a set of E2 nodes (Fig. 1) to which user devices, including IoT devices, are attached. The adversary controls a subset of these devices and uses them to perform flooding, scanning, brute-force login, or web attacks, which are the attack categories present in our corpora. The adversary does not control the RIC, the E2 interface, or the xApp. The defender's xApp consumes either radio telemetry windows delivered through E2SM-KPM or flow records produced by a flow exporter co-located with the user plane, and it may act through E2SM-RC. Since E2 does not carry packets, a flow-level detector requires such an exporter outside the RIC. The cost and placement of this exporter are discussed in Section VII-A.

B. CRITERION 1: GENERALIZATION ACROSS DEPLOYMENTS

Let $\mathcal{D}_A$ and $\mathcal{D}_B$ denote two independently collected corpora represented in a shared feature space. A model f_θ trained on $\mathcal{D}_A$ is characterized by the difference between its held-out score on $\mathcal{D}_A$ and its score on $\mathcal{D}_B$. Because the two corpora differ in attack prevalence (Table 2), a difference in macro- F_1 combines a change in the classifier's behavior with a change in class balance; indeed, even a classifier that ignores its input obtains a different macro- F_1 when the prevalence changes. We therefore report the difference in balanced accuracy, which is invariant to changes in prevalence,

$$\Delta_{BA} = \text{BA}(f_\theta; \mathcal{D}_A^{\text{test}}) - \text{BA}(f_\theta; \mathcal{D}_B), \quad (1)$$

together with the macro- F_1 gap Δ_{F_1} and its difference from the corresponding gap of the majority-class baseline. Covariate shift is quantified feature by feature as

$$\bar{w}_1 = \frac{1}{|\mathcal{F}|} \sum_{j \in \mathcal{F}} w_1(Q_j(P_A^{(j)}), Q_j(P_B^{(j)})), \quad (2)$$

where Q_j denotes the quantile transform of feature j , fitted on the source training data only. If each corpus were normalized by its own quantiles, both marginals would be mapped to the uniform distribution, and every term would be zero.

C. CRITERION 2: ALERT BURDEN UNDER A REALISTIC BASE RATE

Let λ_b denote the benign arrival rate in the detector's own unit, that is, flow records per hour for a flow-level detector and telemetry windows per hour for a window-level detector. A benign UE generates 225 windows of 16 s per hour. The number of false alerts per hour is

$$A(\tau) = \lambda_b \text{FPR}(\tau), \quad (3)$$

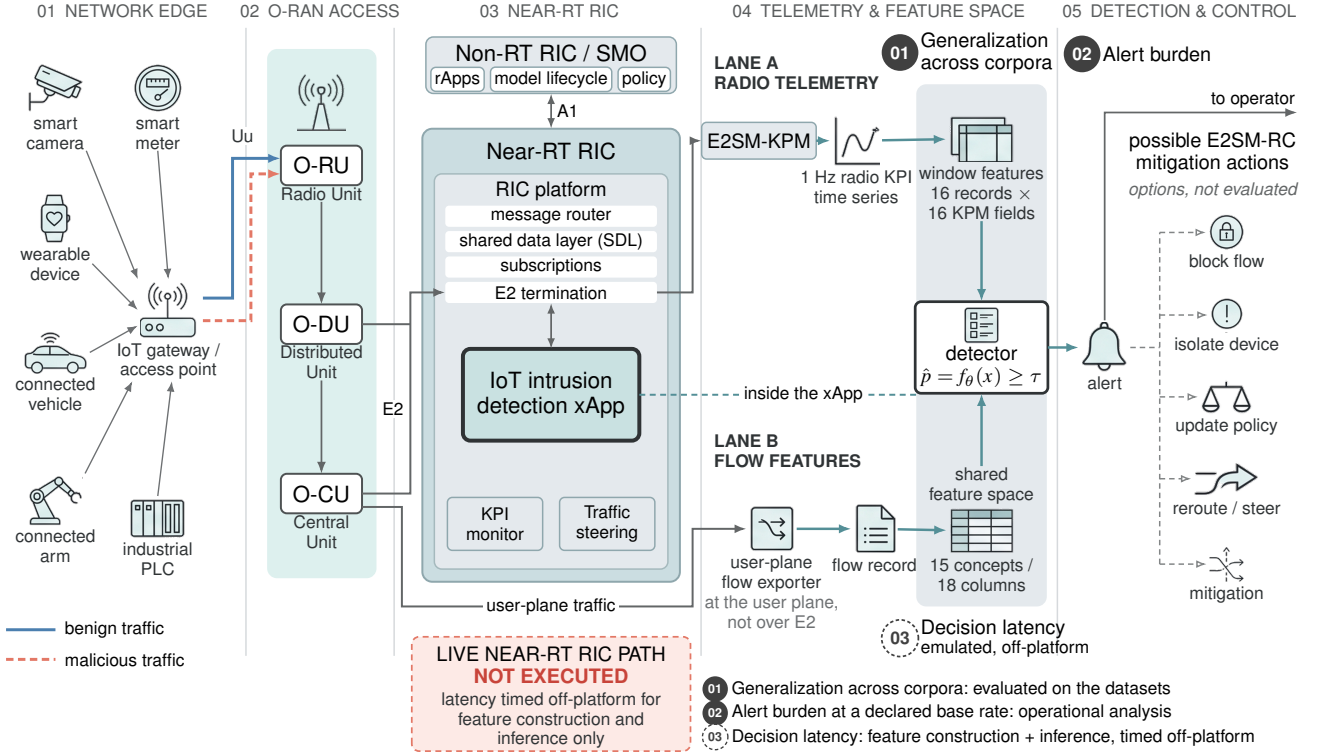


FIGURE 1. Reference O-RAN edge deployment and the detection loop. All device traffic reaches the radio unit over Uu. The detection xApp receives radio telemetry through the E2 termination (lane A) and flow records from an exporter at the user plane, mapped onto the shared feature space (lane B); column 04 expands the xApp's own feature construction and inference, with one detector per layer. Numbered markers show what this paper measures: (01) generalization across corpora and (02) alert burden are evaluated on the corpora; (03) decision latency is timed off-platform on one host, with no RIC in the path. The mitigation actions on the right are options E2SM-RC offers and are not evaluated.

and, for an operational attack prevalence π , the fraction of alerts that correspond to attacks is

$$\text{PPV}(\tau) = \frac{\pi R(\tau)}{\pi R(\tau) + (1 - \pi) \text{FPR}(\tau)}. \quad (4)$$

Neither π nor λ_b can be measured from the corpora, and both are declared and varied over a range. Because PPV depends only on π and A scales linearly with λ_b , we also report the number of false alerts per 10^6 benign flows, which is independent of λ_b .

D. CRITERION 3: LATENCY AGAINST THE CONTROL BUDGET

The decision latency of an xApp can be decomposed as

$$L = t_{\text{ind}} + t_{\text{feat}} + t_{\text{inf}} + t_{\text{act}} + t_q, \quad (5)$$

where the terms correspond to indication decoding, feature construction, inference, control encoding, and queueing. Only t_{feat} and t_{inf} are measured in this study, because the remaining three terms require a RIC in the measurement path. Conformance is a property of the tail of the latency distribution: for a budget B and a tolerated violation rate ε , we require $q_{1-\varepsilon}(L) \leq B$. Since the O-RAN E2 specification allows near-real-time latency requirements anywhere between 10 ms and 1 s [17], we report the smallest budget that is met rather than a verdict for a single chosen value.

E. A DEPLOYABILITY TEST

The three criteria are combined into a test that a detector either passes or fails:

$$\begin{aligned} \mathcal{P}(f_\theta) = & \mathbf{1}[\bar{\Delta}_{\text{BA}} \leq \delta] \cdot \mathbf{1}[\bar{q}_{0.99}(L) \leq B] \\ & \cdot \mathbf{1}[\exists \tau : R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho], \end{aligned} \quad (6)$$

where $\bar{\Delta}_{\text{BA}}$ and $\bar{q}_{0.99}$ denote upper 95% confidence bounds, so that a pass cannot depend on a favorable point estimate, and all alert conditions are evaluated on the target corpus. The minimum recall $r_{\min}$ prevents a detector that never raises an alert from satisfying the alert term. The operating threshold is chosen as the most sensitive admissible threshold,

$$\begin{aligned} \tau^* = \arg \max_{\tau} R(\tau) \quad \text{s.t.} \quad & R(\tau) \geq r_{\min}, A(\tau) \leq A_{\max}, \\ & \text{PPV}(\tau) \geq \rho. \end{aligned} \quad (7)$$

We propose this test as a reporting instrument rather than as an O-RAN requirement. Its bounds represent operator requirements, and Section VII-B reports the values used in this study and the sensitivity of the verdict to them.

IV. METHODOLOGY

This section describes the corpora and the feature spaces, the detectors, the evaluation protocols, and the statistical analysis.

A. CORPORA

$\mathcal{D}_A$ is NetsLab-5GORAN-IDD [39], which was captured on a physical OpenAirInterface O-RAN testbed and contains benign traffic and five attack categories: denial of service (DoS), distributed denial of service (DDoS), probing, brute force, and web attacks. The corpus provides two layers that cannot be joined record by record from the published files, namely flow records exported by Zeek at the central unit and radio telemetry sampled at 1 Hz at the distributed unit. Both layers are used in this study. The radio layer is aggregated into windows of 16 consecutive records of one UE, each summarized by the mean and standard deviation of 16 KPM fields. Its 30 capture sessions are recovered from gaps of more than five minutes in the timestamps. Fard et al. describe the same corpus as 42 experiment runs [31]. A gap threshold of 90 to 105 s recovers exactly 42 segments, each with a single attack category, and each of our sessions is a union of complete segments. The five-minute rule merges runs that were recorded consecutively (10 of the 30 sessions contain two or three attack subcategories), which ensures that such runs always fall on the same side of a split. Each recovered session carries the same category label in every record, so that a radio label is effectively a session label and every window of an attack session is an attack window. The sessions follow the capture schedule (Section VI-B), with nine benign sessions on days 0 to 6, the attack categories recorded in blocks on days 47 to 53, and one final benign session. $\mathcal{D}_B$ is 5G-NIDD [37], [38], which was captured on a 5G test network at a different institution and contains flow records exported by Argus with two attack categories, DoS and probing. Its published files contain no addresses, ports, or timestamps, but they retain the Argus record counter and file offset. Because the offset returns to its initial value at every file boundary, 20 capture files can be recovered. These correspond to two passes over the same ten captures, one per base station, and the published `Combined.csv` also restarts its row index at the beginning of the second pass. Each attack file contains benign traffic together with one attack type. A third corpus, $\mathcal{D}_C$, serves as a second transfer target from a third site. It consists of the flow records of a 5G core testbed (Open5GS in Docker) published by Nugraha et al. [22]. Only three of its files contain flow records, each with one attack (a SYN flood, an ICMP flood, and a PFCP session-deletion attack), for a total of 39,425 flows, of which 839 are attacks. Its benign flows comprise user traffic and the signalling of the core network itself. $\mathcal{D}_C$ was exported by NFStream, a third flow exporter, which counts link-layer bytes and, for flows inside the GTP-U user-plane tunnel, the 44 bytes of encapsulation. We remove both overheads per packet so that the shared columns contain the IP bytes of the inner packet, as they do for $\mathcal{D}_A$. Table 2 summarizes the datasets.

B. SHARED FEATURE SPACE

Because Zeek and Argus share no column names, the flow feature space is defined by the meaning of each field (Table 3). It comprises 15 concepts realized as 18 columns,

TABLE 2. Datasets. $\mathcal{D}_A$ supplies training and in-distribution testing on both layers. $\mathcal{D}_B$ and $\mathcal{D}_C$ are transfer targets; each also trains its own in-target references, and $\mathcal{D}_B$ trains the reverse direction.

	$\mathcal{D}_A$ network	$\mathcal{D}_A$ radio	$\mathcal{D}_B$	$\mathcal{D}_C$
Unit	flow	16 s UE window	flow	flow
Source	Zeek	KPM, 1 Hz	Argus	NFStream
Samples	1 640 182	2 808	1 215 889	39 425
Features	18	32	18	18
Attack prevalence	94.63%	76.46%	60.71%	2.13%
Attack categories	5	5	2	3
Groups	318 hosts	30 runs	20 files	3 files
Role	train, test	train, test	target	target

TABLE 3. Shared flow feature space, generated from `configs/features/shared_space.yaml`. The exporter-robust subset is the protocol one-hot, the three mean packet sizes, and the two direction ratios.

Concept	$\mathcal{D}_A$ (Zeek)	$\mathcal{D}_B$ (Argus)	Role
duration	duration	Dur	base
src_bytes	src_ip_bytes	SrcBytes	base
dst_bytes	dst_ip_bytes	DstBytes	base
src_pkts	src_pkts	SrcPkts	base
dst_pkts	dst_pkts	DstPkts	base
tot_pkts	src_pkts + dst_pkts	TotPkts	base
tot_bytes	src_ip_bytes + dst_ip_bytes	TotBytes	base
proto	proto	Proto	one-hot (4)
mean_pkt_size	tot_bytes / max(tot_pkts, 1)		derived
src_mean_pkt_size	src_bytes / max(src_pkts, 1)		derived
dst_mean_pkt_size	dst_bytes / max(dst_pkts, 1)		derived
bytes_per_s	tot_bytes / max(duration, duration_floor_s)		derived
pkts_per_s	tot_pkts / max(duration, duration_floor_s)		derived
src_byte_ratio	src_bytes / max(tot_bytes, 1)		derived
src_pkt_ratio	src_pkts / max(tot_pkts, 1)		derived

namely seven counts reported by both exporters, a one-hot encoded protocol with a fixed list of levels, and seven derived quantities computed identically for both corpora. One mapping requires particular care. Zeek's `src_bytes` counts application payload, which is zero for most floods and scans, whereas Argus's `SrcBytes` counts complete packets. The correct counterpart of `SrcBytes` is Zeek's `src_ip_bytes`, with 40 and 42 bytes per packet, respectively. A lower bound on duration and the transform $\log(1+x)$ for unbounded quantities are applied identically to both corpora, and scaling is fitted on the source training data only. No address, port, or timestamp is used as a feature.

Before any result was computed with it, we also defined an exporter-robust subset of nine columns, consisting of the protocol and the five per-packet means and direction ratios. The two exporters differ in where they terminate a flow, which affects duration, rates, and totals. Per-packet means and direction ratios are less affected, but they are not invariant. Splitting a flow changes them whenever its packets differ, for example when handshake and teardown packets are assigned to different records, and the two exporters count the same packet as 40 and 42 bytes, which corresponds to a fixed offset of about 5% for minimum-size packets.

TABLE 4. Hyperparameters, identical across protocols and seeds.

Model	Settings
Majority, Strat.	training majority; samples of the training prior
LR	<code>max_iter</code> 2000, balanced weights, standardized
DT	depth 12, balanced weights
RF	200 trees, min. leaf 2, balanced subsample weights
XGB	300 trees, depth 6, $\eta = 0.1$, subsample 0.8, <code>hist</code>
HGB	300 iterations, depth 6, $\eta = 0.1$, balanced weights
MLP	128–64, Adam, early stopping (patience 10), balanced sample weights, standardized
GRU	hidden 32; BCE with positive weight n_-/n_+
1D-CNN	two convolutions (32, kernel 3), average pooling
IF	200 trees; benign-only training
AE	12–6–12 MLP, reconstruction error; benign-only
CORAL	per-domain z-score, identity-regularized covariances

C. DETECTORS

Six supervised architectures and two trivial baselines are evaluated under every protocol. The supervised models are LR, a DT of depth 12, an RF [56], XGB [57], the HGB implementation of scikit-learn [58], and an MLP, and they are compared with a majority-class predictor and a stratified sampler of the training prior. Because the baselines ignore their input, any change in their score between two protocols reflects a change in class composition alone. All supervised models use the same policy for class imbalance, namely balanced class weights, which are applied as sample weights for the MLP and through `scale_pos_weight` for XGB. The hyperparameters are fixed and were not tuned (Table 4). Tuning on held-out source data would favor structure that is specific to the source, which is precisely the effect that the transfer experiments measure, and tuning on the target would leak target information into training.

Three further model families are used to address specific questions. Two sequence models, a GRU [59] and a one-dimensional convolutional network, operate on the raw 16-step KPM sequences of the same radio windows. Two unsupervised detectors, an isolation forest [60] and an autoencoder, are trained on benign source flows only, and their alarm threshold is fixed at the 99th percentile of the scores of benign source flows. Finally, a CORAL variant [54] re-colors the source flows to the covariance of the unlabeled target before training. This is the only experiment in which a detector is exposed to target data.

D. EVALUATION PROTOCOLS

Splits are group-disjoint wherever a group key is available, and an automated check rejects any split in which a group appears on both sides. On the radio layer, the group key is the capture session, and three protocols are compared: a random split of windows, a run-disjoint split, and a category-stratified run-disjoint split that keeps sessions of every attack category on both sides, so that no test category is absent from training. Two further designs address the effect of time (Section VI-B). In the first, the latest or the earliest session of every category is held out and compared with draws that hold out a random

session of every category. In the second, each benign session is held out individually. On the flow layer, the group key is the source address (318 groups). For each of 20 split seeds, a subsample of 300,000 flows is drawn from $\mathcal{D}_A$ and divided by source address. Every model is then evaluated on the whole of $\mathcal{D}_B$, whereas the reverse direction trains on a random split of $\mathcal{D}_B$ and evaluates on $\mathcal{D}_A$. Within $\mathcal{D}_B$ itself, the in-target references either hold out four of the 20 capture files or train on the pass of one base station and test on the pass of the other (Section VI-C). Apart from these references, CORAL, the domain classifier, and the prior estimates of Section VI-H, all of which are identified as such, no transform, threshold, or model choice uses $\mathcal{D}_B$, and every access to it is logged.

E. STATISTICAL ANALYSIS

Uncertainty is estimated over 20 split seeds. Because the splits resample a single population of 30 sessions or 318 source addresses, their training and test sets overlap, and a paired *t*-test over split means underestimates the variance. We therefore report Nadeau–Bengio corrected intervals [55], which scale the variance by $1/J + n_{\text{test}}/n_{\text{train}}$ for J splits. Holm correction is applied within each family of tests, where a family consists of one metric and one comparison (a protocol pair or a transfer direction) across the six supervised architectures, and the two sequence models form a separate family. Operating points are computed from confusion counts pooled over folds, and Section VI-G explains why averaging per-fold precision is inappropriate. Pooled counts include a sample once for every seed in which it is tested, and samples within a cluster are correlated, so a binomial interval computed from them would be much too narrow. Intervals for pooled operating points are instead obtained from a cluster bootstrap with 2,000 replicates, which resamples with replacement both the split seeds and the clusters, that is, radio sessions, $\mathcal{D}_A$ source addresses, or $\mathcal{D}_B$ capture files.

F. EXPERIMENTAL PIPELINE AND INFERENCE LOOP

Fig. 2 summarizes the experimental pipeline, and Algorithm 1 describes the decision loop that a detection xApp would execute. The algorithm specifies the intended control logic only. It was not deployed in a RIC, and the latency measurements in Section VI-I cover its feature construction and inference stages alone. A decision that exceeds the budget is still acted upon and recorded as a violation, since a late mitigation is generally preferable to none. An operator who prefers to discard late actions can move the transmission step inside the deadline check.

V. EXPERIMENTAL SETUP

All measurements were performed on a single workstation with a 13th Gen Intel Core i9-13900H processor (14 cores, 20 threads), 47.6 GB of memory, and Windows 11 Pro build 26120, using Python 3.11.9, scikit-learn 1.8.0, XGBoost 3.2.0, NumPy 2.3.5, and ONNX Runtime 1.30.0. The sequence models were trained with PyTorch 2.14 on the CPU in a separate environment. A complete list of packages is

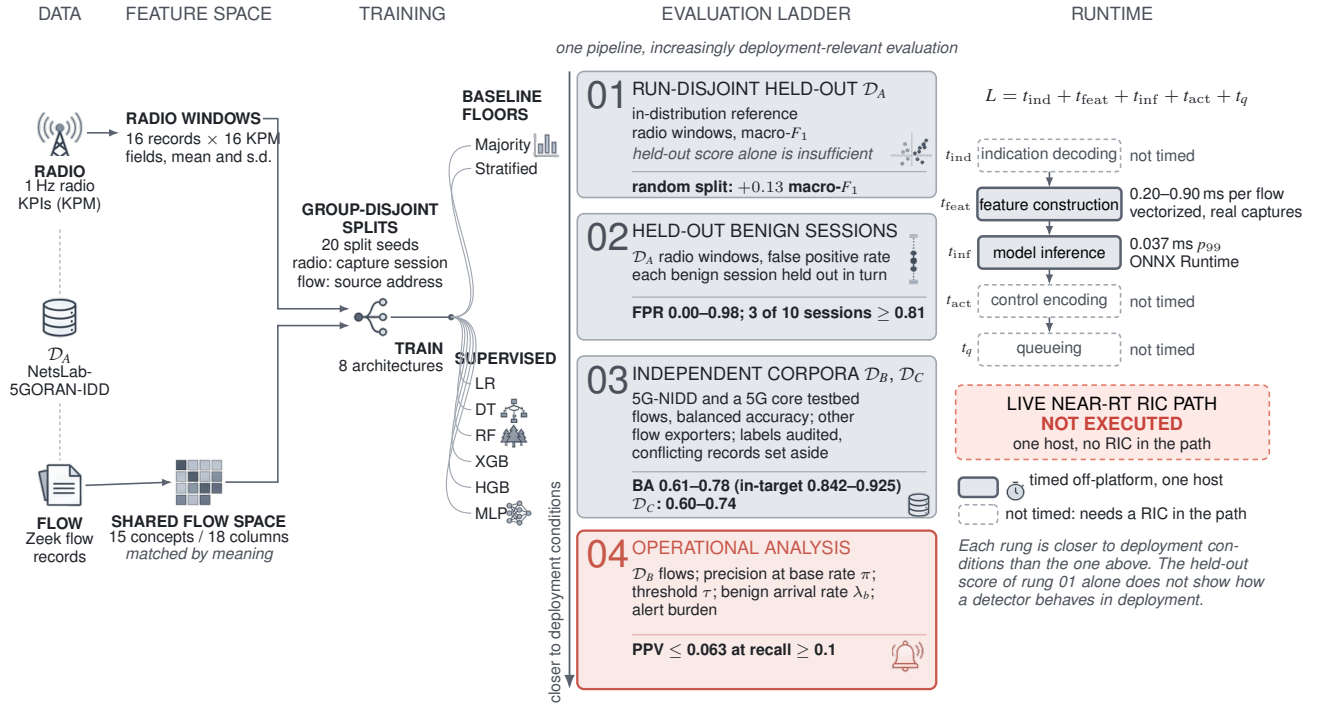


FIGURE 2. Experimental pipeline and evaluation ladder. Radio windows and flows from $\mathcal{D}_A$ are summarized separately, split by group, and used to train eight architectures. Each rung of the ladder moves closer to deployment conditions: a run-disjoint held-out split, held-out benign sessions, two independently collected corpora evaluated on flows with consistent labels, and an operational analysis at a declared base rate. The bold line under each rung gives the measured result. In the runtime column only feature construction and inference were timed, on one host with no RIC in the path; the live-RIC arm was not executed.

Algorithm 1 Budget-aware decision loop of a detection xApp

Require: indication stream $\mathcal{S}$, model f_θ , threshold τ , budget B , margin δ_m

Ensure: E2 control actions; violation count v ; latency record

```

1:  $\mathcal{W} \leftarrow$  empty per-UE window map;  $v \leftarrow 0$ 
2: for all indication  $m$  arriving on  $\mathcal{S}$  do
3:    $t_0 \leftarrow \text{Now}()$ ;  $u \leftarrow \text{UEOf}(m)$ 
4:    $\mathcal{W}[u] \leftarrow \text{UpdateWindow}(\mathcal{W}[u], m)$ 
5:    $x \leftarrow \text{Features}(\mathcal{W}[u])$ 
6:    $\hat{p} \leftarrow f_\theta(x)$ 
7:   if  $\hat{p} \geq \tau$  then
8:     SendE2(EncodeControl( $u, \hat{p}$ ))  $\triangleright$  even if late
9:   end if
10:   $\ell \leftarrow \text{Now}() - t_0$ 
11:  if  $\ell > B - \delta_m$  then
12:     $v \leftarrow v + 1$   $\triangleright$  deadline violation, recorded
13:  end if
14:  RecordLatency( $\ell$ )  $\triangleright$  every indication
15: end for

```

provided with the results. No RIC was involved in any experiment. The detector was not deployed as an xApp, no E2 agent was run, no component was containerized, and all latency figures are therefore emulated.

The comparison of protocols on the radio layer uses 20 split seeds with two model seeds each, and the remaining radio-layer and flow-layer experiments use 20 split seeds with one

model seed. Calibration uses 10 split seeds of a three-way source split into training, calibration, and test groups. The declared operational parameters are an attack prevalence of $\pi = 0.002$ and a benign arrival rate of $\lambda_b = 240,000$ flows per hour, and both are varied over a range. For the radio layer, false alerts are reported per benign UE-hour. Neither parameter was obtained from an operator. The chosen rate of about 67 flows per second is below the several hundred flows per second that Jin et al. expect at a heavily loaded base station of a large LTE network [61], so the reported alert volumes are likely to underestimate those of an edge site serving several cells. Latency is measured per call with a monotonic clock, with garbage collection enabled, after 1,000 warm-up calls, over 20,000 calls, and with one thread per library. Distribution-free confidence intervals are reported for every quantile.

VI. RESULTS

The results are presented in the order of the evaluation ladder. We first examine protocol sensitivity within one corpus, followed by the effects of time order and benign sessions, the properties of the target corpus and of a published result on it, transfer across corpora and the composition of the transfer gap, and a third corpus. We then report the alert burden, calibration, and latency.

TABLE 5. Protocol sensitivity on the $\mathcal{D}_A$ radio layer: macro- F_1 under random, run-disjoint (RD) and category-stratified run-disjoint (CS) splits, and paired gains of the random split with Nadeau-Bengio corrected 95% intervals over 20 split seeds. No supervised architecture's gain is significant after Holm correction. 1-NN: a model-free nearest-neighbor lookup on the same splits.

Model	Random	Run-disj.	Cat.-strat.	Δ_{R-RD} [95% CI]	Δ_{R-CS} [95% CI]
LR	0.836	0.744	0.697	+0.093 [-0.05, +0.24]	+0.139 [-0.06, +0.33]
DT	0.950	0.782	0.758	+0.168 [+0.02, +0.32]	+0.192 [+0.02, +0.36]
RF	0.970	0.853	0.816	+0.117 [-0.04, +0.27]	+0.154 [-0.02, +0.32]
XGB	0.977	0.837	0.808	+0.140 [-0.00, +0.28]	+0.169 [-0.02, +0.35]
HGB	0.982	0.844	0.831	+0.138 [-0.01, +0.28]	+0.152 [-0.03, +0.33]
MLP	0.954	0.828	0.809	+0.126 [-0.01, +0.26]	+0.146 [-0.02, +0.31]
Strat.	0.498	0.507	0.500	-0.009 [-0.01, -0.00]	-0.002 [-0.01, +0.01]
Maj.	0.431	0.434	0.447	-0.002 [-0.02, +0.01]	-0.016 [-0.04, +0.01]
1-NN	0.958	0.814	0.794	+0.144 [+0.00, +0.28]	+0.164 [+0.01, +0.31]

A. EVALUATION-PROTOCOL SENSITIVITY

Table 5 and Fig. 3 compare three protocols on the radio layer. Relative to a run-disjoint split, a random split raises macro- F_1 by 0.09 to 0.17 across the six architectures, or by 0.13 on average, and the gain is positive in at least 16 of the 20 splits for every architecture. The trivial baselines, in contrast, show no gain. The corrected interval for the average gain is [-0.01, 0.27] ($p = 0.06$). With only 30 capture sessions, the magnitude of the effect cannot be estimated precisely, and the gain of no individual architecture remains significant after Holm correction. For comparison, an uncorrected paired t -test would yield $p = 6 \times 10^{-5}$ for the same data.

The gain cannot be attributed to test categories that are absent from training, since every test category also appeared in training in all 20 run-disjoint splits. The category-stratified protocol guarantees this property by construction and yields a gain of 0.14 to 0.19. However, this protocol moves sessions of rare categories into the test set, which makes its test sets more difficult, and we do not interpret its larger gain as additional evidence. A model-free probe helps to identify the underlying mechanism. Under a random split, 68% of the test windows have their nearest training window in the same capture session (65% to 71% across splits), and a one-nearest-neighbor (1-NN) lookup achieves a macro- F_1 of 0.96, compared with 0.81 under a run-disjoint split (Table 5). Because every radio label is also a session label, recognizing the session is sufficient to predict the label, and a random split allows any model, including a lookup table, to exploit this. Under the run-disjoint split, the lookup outperforms LR and DT and comes within 0.04 of the best supervised architecture, which suggests that, on this layer, the learned models add little beyond the retrieval of similar windows. The two sequence models in Table 6, which operate on the raw 16-step KPM sequences instead of window summaries, exhibit the same pattern more clearly. They achieve 0.923 to 0.944 under a random split and 0.755 to 0.759 under a run-disjoint split, a gain of 0.16 to 0.19 whose corrected intervals exclude zero for both models (Holm-adjusted $p \leq 0.02$ within the pair).

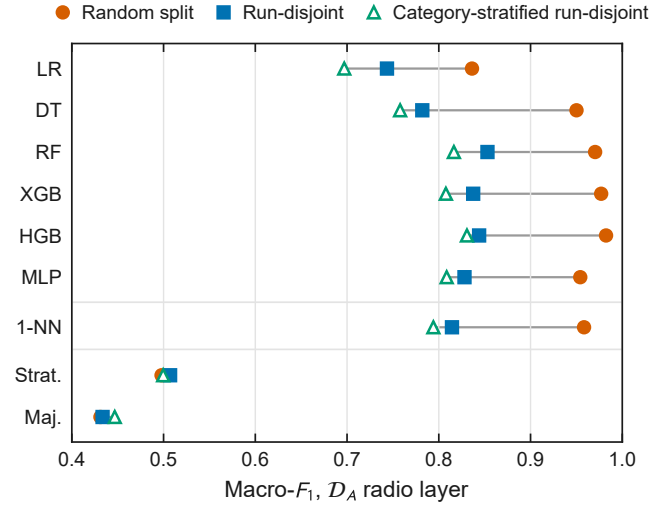


FIGURE 3. Macro- F_1 on the $\mathcal{D}_A$ radio layer under three protocols, mean over 20 split seeds; the gray line spans the three protocols of each model. The random split increases the score of every supervised model and of the model-free 1-NN lookup, whereas the scores of the input-blind baselines remain unchanged. Retaining every attack category in training (category-stratified split) does not eliminate the decrease. Corrected intervals are given in Table 5.

TABLE 6. Sequence models on the same radio windows and splits as Table 5.

Model	Random	Run-disj.	Cat.-strat.	Δ_{R-RD} [95% CI]
1D-CNN	0.944	0.755	0.763	+0.189 [+0.05, +0.32]
GRU	0.923	0.759	0.769	+0.165 [+0.04, +0.29]

B. TIME ORDER AND BENIGN SESSIONS

Although the radio layer spans 52.96 days, its capture order follows the scenario schedule (Table 7). Nine benign sessions were recorded on days 0 to 6, the five attack categories were recorded in blocks on days 47 to 53, and a final benign session was recorded on day 53. Because every radio label is a session label, each session is either benign or belongs to a single attack category throughout. A split based on session order consequently holds out entire attack categories. For training fractions between 0.4 and 0.7, the latest sessions contain 1 to 4 attack categories that are absent from training, together with a single benign session. Such a split measures the detection of unseen categories rather than temporal drift, and we therefore do not report it as a drift estimate.

To keep category coverage fixed, we held out the latest session of every category, including the benign category, and compared the result with 50 control draws, each of which holds out a randomly chosen session of every category (Table 8). When the latest sessions are held out, balanced accuracy ranges from 0.52 to 0.57, compared with control means of 0.75 to 0.83, and yet 12% to 32% of the control draws score equally low. When the earliest sessions are held out, balanced accuracy ranges from 0.95 to 0.99, a level that 4% to 34% of the control draws match or exceed. The effect of time order thus remains within the ordinary variation between sessions.

The variation is determined largely by which benign ses-

TABLE 7. False positive rate on each benign radio session when that session alone is held out and the models train on all other 29 sessions, mean and range over the six architectures. Day: capture start relative to the first session. Alerts per benign UE-hour: $FPR \times 225$ windows.

Session	Day	UEs	Windows	FPR	Range	Alerts/UE-h
0	0.0	1	57	0.00	[0.00, 0.00]	0
1	0.0	1	54	0.02	[0.00, 0.06]	6
2	2.8	1	58	0.00	[0.00, 0.00]	0
3	2.9	1	55	0.07	[0.02, 0.20]	15
4	3.0	1	24	0.81	[0.54, 0.96]	181
5	3.9	1	56	0.14	[0.00, 0.77]	31
6	4.0	2	60	0.00	[0.00, 0.00]	0
7	4.0	1	95	0.01	[0.00, 0.03]	2
8	6.0	1	56	0.98	[0.93, 1.00]	220
29	52.9	3	146	0.87	[0.84, 0.93]	196

TABLE 8. Category-stratified time split on the $\mathcal{D}_A$ radio layer: balanced accuracy when the latest or the earliest session of every category is held out, against the mean of 50 draws that hold out a random session of every category. “% ctrl. $\leq$ ”: share of control draws scoring at or below the time split. Every test category is in training.

Model	Control		Latest held out		Earliest held out	
	BA	% ctrl. $\leq$	BA	% ctrl. $\leq$	BA	% ctrl. $\leq$
LR	0.749	0.573	0.950	96	0.950	96
DT	0.774	0.521	0.962	84	0.962	84
RF	0.811	0.562	0.970	66	0.970	66
XGB	0.812	0.567	0.978	72	0.978	72
HGB	0.817	0.550	0.995	94	0.995	94
MLP	0.827	0.574	0.983	94	0.983	94

sion is placed in the test set. When the benign sessions are held out one at a time (Table 7 and Fig. 4), 6 of the ten sessions yield a false positive rate of at most 0.07, whereas sessions 4, 8 and 29 (days 3, 6 and 53) yield rates between 0.81 and 0.98. The final benign session, the only one recorded during the attack period, yields a false positive rate of 0.87, which corresponds to 196 false alerts per benign UE-hour, but two sessions from the first week yield comparable rates. With only ten benign sessions, a change over time cannot be separated from this heterogeneity. In practical terms, the false positive rate of a detector at a new site cannot be inferred from its rate on the benign sessions on which it was tested.

C. THE TARGET CORPUS: LABELS AND A PUBLISHED RESULT

Label conflicts. Before evaluating transfer to $\mathcal{D}_B$, we examined whether its labels are a function of its flow records, and found that this is not the case for half of the corpus. Of the 1,215,889 flows, 52% (632,378) share every field except the row index, Seq, and Offset with a flow that carries the opposite label (Table 9, Fig. 5). All of these flows lie in the two UDP-flood captures, where they account for 75% of file 5 and 98% of file 15, one file from each base station. The pattern is highly specific. Each of the 281,529 flows labeled as UDP flood in file 15 has an identical copy labeled as benign in file 5, and the counts are equal for 33,704 of the 33,708 distinct records. The same traffic appears to have been captured at both stations and labeled differently at each. These copies constitute 59% of the benign class of the corpus. Because the published files do not indicate which

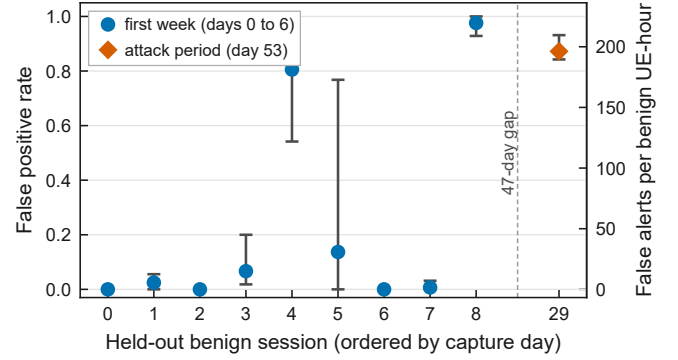


FIGURE 4. False positive rate on each benign radio session when that session alone is held out: mean (marker) and range (bar) over the six architectures, sessions in capture order. The right axis gives the same rate as false alerts per benign UE-hour (225 windows). Sessions recorded in the same week range from no false positives to almost all windows flagged, which indicates that the false positive rate depends more on the particular benign session than on the time of capture.

label is correct, every $\mathcal{D}_B$ result below is reported both with and without these benign copies, and no label is modified.

These conflicts impose an upper bound on the performance of any classifier that uses the record content. Even when evaluated on the flows from which it was constructed, the best possible function of the 90 content fields achieves a balanced accuracy of 0.766. In the 18 shared columns the corresponding bound is 0.753, which is the level reached by DT, XGB, HGB, and the MLP when they are trained and tested on a random split of $\mathcal{D}_B$ (Table 10). When Seq and Offset are added, every record becomes unique and the bound rises to 1.000.

References within the target. To distinguish the cost of transfer from the limits imposed by $\mathcal{D}_B$ itself, we also trained and tested the six architectures on $\mathcal{D}_B$ alone (Table 10). On a random split, they reach 0.708 to 0.752 in the shared space, which corresponds to the bound given above, and 0.842 to 0.925 when the benign copies are excluded from the test set. On held-out capture files, they reach 0.931 to 0.982 in the shared space and 0.990 to 1.000 with native features, although none of the three draws held out a UDP-flood capture. Across the two base stations, within a single corpus and with a single exporter, balanced accuracy decreases to 0.618 to 0.641 from station 1 to station 2 and to 0.639 to 0.650 in the opposite direction. Both decreases are caused by the copies. A model trained on station 1, where the flood copies are labeled benign, misses the floods of station 2, whereas a model trained on station 2 flags the benign-labeled copies of station 1 and scores 0.973 to 0.988 when these copies are removed from the test set. On flows whose labels are determined by the record content, the shared space separates the classes of $\mathcal{D}_B$ almost perfectly, which indicates that the target corpus itself is not the factor that limits transfer.

A published result under the same ladder. The dataset authors report binary accuracies of 99.85% to 99.95% on a random 70/30 split [11]. We reproduced their pipeline, which consists of the published one-hot encoding, selection

TABLE 9. Label conflicts in $\mathcal{D}_B$ (5G-NIDD). Content: all fields of the published encoding except row index, Seq, Offset, and the labels. Ceiling: balanced accuracy of the best lookup on distinct records, in sample.

Features	Cols	Records	% in conflicting records			BA ceiling
			All	Benign	Attack	
Content fields	90	374 946	52.0	58.9	47.5	0.766
Shared space	18	246 281	62.4	59.7	64.1	0.753
Content + Seq, Offset	92	1 215 869	0.0	0.0	0.0	1.000

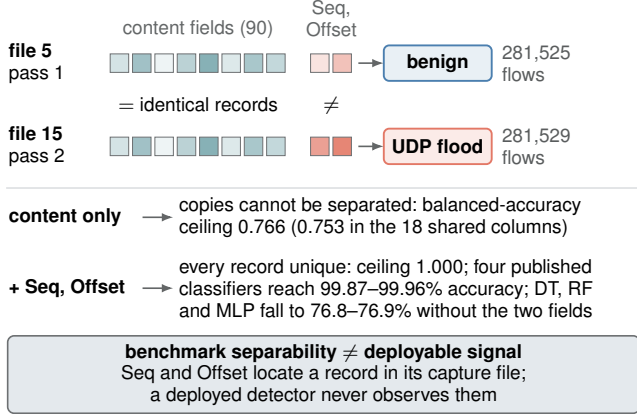


FIGURE 5. Illustration of the label conflict in $\mathcal{D}_B$. Identical flow records occur in two capture files from the two passes and are labeled benign in one file and UDP flood in the other (the tiles are schematic). Because no function of the record content can separate the copies, the attainable balanced accuracy is bounded. The record-position fields Seq and Offset differ between the copies, make every record unique, and account for the published accuracy of 99.9%, although a deployed detector cannot observe them.

of the ten features with the highest ANOVA F -scores, z -score normalization, and their five classifiers, for which we used the scikit-learn default settings because the tuned values are not reported. Our F -scores rank the features in exactly the same order, and on their split we obtain accuracies of 99.87% to 99.96% for DT, RF, KNN, and the MLP (Naive Bayes 92.71%), in agreement with their results (Table 11). Their first and second features are Seq and Offset, which encode the position of a record within its capture file. A deployed detector cannot observe these fields, and, according to the bounds given above, they are what separates the conflicting records. Replacing only these two fields with the next two features in their ranking reduces the accuracy of DT, RF, and the MLP to 76.8% to 76.9% (balanced accuracy 0.704 to 0.706, false positive rate 0.588 to 0.591). This decrease reflects the label conflict rather than a loss of attack information, since RF still achieves 0.9999 on the test flows without the benign copies. The two counters thus allow the published models to distinguish identical records by their position within the file. When capture files are held out, the published features yield average balanced accuracies of 0.859 to 0.923 for DT, RF, and the MLP, with RF ranging from 0.82 to 1.00 across folds. Without the counters, the average is 0.799 to 0.800, RF ranges from 0.50 to 1.00, and accuracy falls to 1.6% on the fold that tests one flood capture after training on the other.

Across base stations, the published features yield 0.762 to 0.895, and the features without counters yield 0.656 to 0.692. A published accuracy of 99.9% on this corpus is of limited value for predicting performance on a new capture file or a new base station, and it provides no information about a deployment, in which record positions do not exist.

D. TRANSFER ACROSS CORPORA

Table 12 and Fig. 6 report the results of transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$. In terms of macro- F_1 , every supervised model loses 0.10 to 0.13. However, the majority baseline, which uses no features, also loses 0.11, because $\mathcal{D}_B$ has a lower attack prevalence than $\mathcal{D}_A$. Relative to this baseline, the loss in macro- F_1 lies between -0.018 and +0.015, and none of the macro- F_1 losses remains significant after correction. Macro- F_1 is therefore not an appropriate measure for this comparison. Balanced accuracy, which is unaffected by prevalence, shows the loss clearly. It decreases from 0.81–0.91 on held-out source data to 0.53–0.63 on $\mathcal{D}_B$, a drop of 0.21 to 0.36 that is significant for 5 of the six architectures after correction, is positive in at least 18 of the 20 splits for every architecture, and averages 0.28 [0.15, 0.41]. The trivial baselines remain unchanged, and the target ROC-AUC lies between 0.53 and 0.70.

The detectors remain above chance on the target, although only moderately so. The average target balanced accuracy is 0.57, and the corrected interval excludes 0.5 for 4 of the six architectures (3 after Holm correction). For reference, the stratified sampler achieves a macro- F_1 of 0.424 on $\mathcal{D}_B$, and a fair coin would achieve 0.494. Both values are reported because the value of a macro- F_1 floor depends on the prior from which it samples. According to every prevalence-independent measure, LR transfers best (balanced accuracy 0.63, ROC-AUC 0.70), whereas the MLP, which achieves the highest held-out source balanced accuracy (0.91), reaches 0.55 on the target. Source performance thus does not predict target performance in a prevalence-independent metric, as the rank correlation across the six architectures is -0.26 in balanced accuracy ($p = 0.62$). In macro- F_1 the rank correlation is 0.83 ($p = 0.04$), but rankings in macro- F_1 change with prevalence, and six architectures provide little precision for either estimate.

Effect of the target labels. Part of this loss is attributable to the label conflict described in Section VI-C. Detectors that learned to recognize a UDP flood on $\mathcal{D}_A$ detect the floods in capture file 15 and, by the same rule, flag between 0.00 and 0.85 of the benign-labeled copies in file 5, whereas LR detects neither. When the copies are excluded, in a deterministic re-run of 10 of the 20 split seeds, the target balanced accuracy is 0.61 to 0.78, namely 0.74 to 0.78 for the five nonlinear architectures and 0.61 for LR. The loss relative to held-out source data decreases to 0.00 to 0.25 and is significant for 1 of the six architectures after correction. The conflicts thus reverse the apparent ranking of the detectors. On all flows LR appears to transfer best, whereas on the flows whose labels are determined by the record content it performs worst. Com-

TABLE 10. References inside $\mathcal{D}_B$: balanced accuracy of models trained and tested on $\mathcal{D}_B$ only, mean over n draws of a 240,000-flow training sample. *Range over the six architectures on the test flows without the benign copies of Table 9. Capture-file draws hold out four of the 20 files.

Features	Split	LR	DT	RF	XGB	HGB	MLP	BA*	n
Shared (18)	Random	0.708	0.752	0.723	0.752	0.752	0.750	0.84–0.92	3
Shared (18)	Capture file	0.955	0.982	0.940	0.943	0.956	0.931	0.93–0.98	3
Shared (18)	Station 1 to 2	0.618	0.638	0.640	0.641	0.640	0.635	0.62–0.64	1
Shared (18)	Station 2 to 1	0.639	0.648	0.648	0.648	0.648	0.650	0.97–0.99	1
Native	Random	0.710	0.764	0.735	0.764	0.764	0.764	0.80–0.87	2
Native	Capture file	0.990	1.000	0.999	1.000	1.000	0.994	0.99–1.00	2
Native	Station 1 to 2	0.698	0.661	0.662	0.662	0.662	0.662	0.66–0.70	1
Native	Station 2 to 1	0.648	0.648	0.649	0.650	0.650	0.654	0.99–1.00	1

TABLE 11. The 5G-NIDD authors' pipeline [11] reproduced and moved one step per rung: balanced accuracy on $\mathcal{D}_B$. R0 is their protocol. *RF on the test flows without the benign copies of Table 9. KNN is evaluated at R0 only.

Rung	Split	Features	DT	Balanced accuracy				RF	
				RF	KNN	NB	MLP	BA*	FPR
R0	Random 70/30	Published 10	0.9995	0.9996	0.9988	0.9310	0.9986	0.9997	0.0005
R1	Random 70/30	No Seq/Offset	0.7059	0.7059	–	0.6814	0.7042	0.9999	0.5881
R2	Capture file	Published 10	0.8594	0.9226	–	0.7501	0.8669	0.9578	0.0723
R2	Capture file	No Seq/Offset	0.7988	0.7996	–	0.7593	0.8004	0.8958	0.1937
R3	Base station	Published 10	0.7619	0.8949	–	0.6598	0.8577	0.9354	0.0819
R3	Base station	No Seq/Offset	0.6557	0.6555	–	0.6271	0.6919	0.8284	0.3465

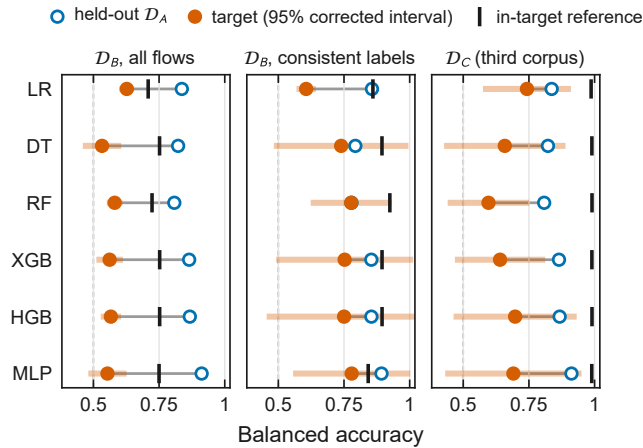


FIGURE 6. Balanced accuracy of detectors trained on $\mathcal{D}_A$: on held-out $\mathcal{D}_A$ (open) and on the target (filled, with the Nadeau-Bengio corrected 95% interval), for $\mathcal{D}_B$ on all flows (20 split seeds), $\mathcal{D}_B$ without the benign copies of Fig. 5 (10 seeds), and the third corpus $\mathcal{D}_C$ (20 seeds). The vertical bar is a reference trained and tested inside the target on a random split. The dashed line is chance, where both input-blind baselines sit. On every target the transferred detectors fall short of the in-target reference.

pared with the in-target reference on these flows (Table 10), transfer still falls short by 0.06 to 0.25 in balanced accuracy.

The reverse direction (Table 13) behaves differently. Balanced accuracy on $\mathcal{D}_A$ reaches 0.61 to 0.76, and the change relative to the source reference, which is optimistic because it is based on a random split, ranges from -0.037 to +0.145. LR, XGB, HGB, and the MLP lose little or no balanced accuracy, whereas DT and RF lose a significant amount. Macro- F_1 nevertheless decreases sharply, because $\mathcal{D}_A$ consists of 94.63% attack traffic and a small number of false negatives dominates the small benign class. A model trained on the narrower $\mathcal{D}_B$

thus separates the classes of $\mathcal{D}_A$ better than a model trained on $\mathcal{D}_A$ separates those of $\mathcal{D}_B$. A possible explanation, which we have not tested, is that the larger and more varied benign traffic of $\mathcal{D}_B$ yields a better model of benign behavior.

Per-category detection (Table 14) shows that no architecture performs acceptably on both classes of $\mathcal{D}_B$. LR retains a benign specificity of 0.98 but detects only 0.25 of the DoS traffic, while the tree models and the MLP detect more attacks at the cost of flagging 0.43 to 0.56 of the benign traffic. In the reverse direction, categories that are absent from $\mathcal{D}_B$ are not detected uniformly. The ensembles and the MLP recall at most 0.01 of the web attacks, whereas brute-force attacks, which are also absent from $\mathcal{D}_B$, are recalled at rates of 0.34 to 0.57.

E. COMPOSITION OF THE TRANSFER GAP

Separability. A classifier trained to distinguish $\mathcal{D}_A$ flows from $\mathcal{D}_B$ flows achieves a balanced accuracy of 0.999 on the 18 shared columns and of 0.997 on the nine exporter-robust columns alone. The largest per-feature shifts concern the protocol mix (most $\mathcal{D}_A$ flows are TCP, whereas most $\mathcal{D}_B$ flows are UDP), the direction ratios, and the packet counts (Fig. 7). Moreover, the robust columns shift as much as the segmentation-dependent columns ($\bar{W}_1$ of 0.28 compared with 0.27). The corpora thus differ in traffic composition, which no choice of exporter would remove, as well as in flow segmentation, which a common exporter would remove.

Model families and adaptation. Table 15 evaluates the same splits under three modifications, none of which yields a transferable detector. Restricting the detectors to the nine exporter-robust columns raises the held-out source balanced accuracy to 0.80 to 0.94, but the target balanced accuracy falls to 0.48 to 0.53, which is no better than chance. This result

TABLE 12. Transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$ in the 18-column shared space, mean over 20 split seeds. Δ_{BA} : source minus target balanced accuracy, with Nadeau–Bengio corrected 95% interval; *: significant after Holm correction. Last column: macro- F_1 loss minus the majority baseline’s loss. Italic rows are the trivial baselines.

Model	Macro- F_1		Balanced accuracy		Δ_{BA} [95% CI]	Target ROC-AUC	Δ_{F_1} vs. Maj.
	Src.	Tgt.	Src.	Tgt.			
LR	0.626	0.530	0.837	0.627	+0.210* [+0.10, +0.32]	0.701	-0.018
DT	0.619	0.518	0.822	0.533	+0.290* [+0.08, +0.50]	0.526	-0.013
RF	0.701	0.572	0.808	0.581	+0.227 [-0.01, +0.47]	0.578	+0.015
XGB	0.646	0.532	0.864	0.562	+0.303* [+0.14, +0.47]	0.572	-0.001
HGB	0.647	0.537	0.867	0.566	+0.301* [+0.13, +0.47]	0.581	-0.004
MLP	0.655	0.531	0.912	0.553	+0.359* [+0.22, +0.50]	0.557	+0.010
<i>Strat.</i>	0.494	0.424	0.502	0.500	+0.001 [-0.01, +0.01]	0.500	-0.044
<i>Maj.</i>	0.492	0.378	0.500	0.500	+0.000 [+0.00, +0.00]	0.500	+0.000

TABLE 13. Reverse direction, $\mathcal{D}_B$ to $\mathcal{D}_A$. The source reference is a random split of $\mathcal{D}_B$, which has no group key, and is therefore optimistic.

Model	Macro- F_1		Balanced accuracy		Δ_{BA} [95% CI]	Target ROC-AUC	Δ_{F_1} vs. Maj.
	Src.	Tgt.	Src.	Tgt.			
LR	0.708	0.563	0.708	0.745	-0.037* [-0.05, -0.02]	0.827	+0.254
DT	0.752	0.343	0.751	0.607	+0.145* [+0.05, +0.24]	0.622	+0.518
RF	0.730	0.337	0.723	0.630	+0.093* [+0.06, +0.13]	0.845	+0.502
XGB	0.753	0.497	0.752	0.764	-0.012 [-0.03, +0.01]	0.845	+0.364
HGB	0.753	0.488	0.752	0.751	+0.001 [-0.05, +0.05]	0.832	+0.373
MLP	0.752	0.470	0.751	0.728	+0.023 [-0.04, +0.09]	0.814	+0.390
<i>Strat.</i>	0.502	0.417	0.502	0.500	+0.003 [+0.00, +0.00]	0.500	+0.194
<i>Maj.</i>	0.377	0.486	0.500	0.500	+0.000 [+0.00, +0.00]	0.500	+0.000

indicates that the columns that do not depend on flow segmentation carry the shift in traffic composition described above. CORAL, which uses unlabeled target features, yields a target balanced accuracy of 0.47 to 0.60. Only the MLP benefits from it (0.60 compared with 0.55 without adaptation), and the target false positive rate increases to 0.55 to 0.99. The novelty detectors trained on benign traffic alone already fail on held-out source data, because floods and scans resemble short benign flows in this feature space.

Cost of the shared space. Projecting $\mathcal{D}_A$ onto the 18 shared columns reduces the in-distribution balanced accuracy by 0.005 to 0.082 relative to the use of all 46 non-port Zeek features (Table 16). The low source macro- F_1 values in Table 12 reflect the 94.63% attack prevalence of $\mathcal{D}_A$ rather than weak detectors, as the corresponding source balanced accuracy is 0.81 to 0.91.

F. A THIRD CORPUS

To examine whether these results depend on the particular pair of corpora, we added $\mathcal{D}_C$, the flow records of a 5G core testbed [22], as a second target. This corpus changes the site, the flow exporter, and the vantage point simultaneously, and, unlike $\mathcal{D}_B$, its labels are consistent. Only 3.2% of its flows lie on records that also occur with the opposite label, and the upper bound on balanced accuracy in the shared columns is 0.992. When trained and tested on $\mathcal{D}_C$ itself, the six architectures achieve 0.987 to 0.990 on a random split and 0.660 to 0.826 when the capture file of one attack type is held out, so that this attack type is not seen during training.

For transfer from $\mathcal{D}_A$ (Table 17, 20 split seeds), the bal-

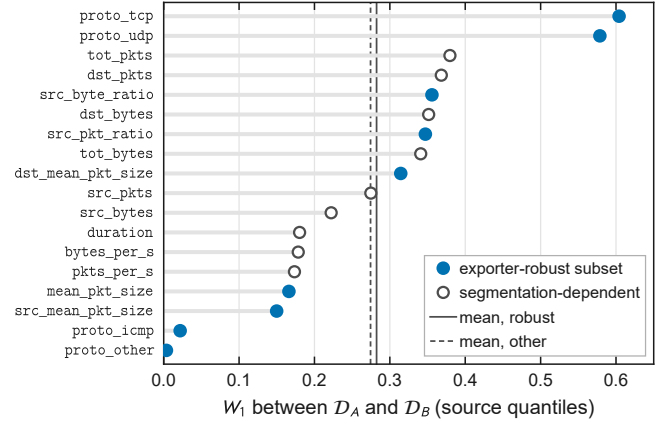


FIGURE 7. Covariate shift between $\mathcal{D}_A$ and $\mathcal{D}_B$ per shared column: first-order Wasserstein distance after the quantile transform fitted on source training data, Eq. (2). Filled markers are the exporter-robust subset, fixed before any result under it was computed; vertical lines are the two group means. The robust columns shift as much as the remaining columns, which indicates that the gap is not solely due to differences in how the two exporters segment flows.

anced accuracy on $\mathcal{D}_C$ ranges from 0.60 to 0.74, and ROC-AUC ranges from 0.70 to 0.93. Every architecture detects the SYN flood, an attack type that is represented in training. Recall on the ICMP flood ranges from 0.10 to 1.00, and recall on the control-plane PFCP attack ranges from 0.00 to 0.99. The main source of error is the benign traffic of the new site. The false positive rate ranges from 0.225 to 0.807, and up to 0.81 of the signalling traffic of the core network is flagged. The loss relative to held-out source data, 0.09 to 0.22, varies so strongly between split seeds that none of the losses is significant after correction. For transfer from $\mathcal{D}_B$ (10 split seeds), balanced accuracy ranges widely from 0.42 to 0.85, and ROC-AUC is as low as 0.38. Which architecture transfers best depends on the pair of corpora, and no architecture transfers consistently. At $\pi = 0.002$, the highest operational precision at any threshold that retains recall above 10% is 0.092 for transfer from $\mathcal{D}_A$ and 0.031 for transfer from $\mathcal{D}_B$, and no architecture passes the deployability test in either direction (0 and 0 of six). The pattern observed in Section VI-D thus recurs at a third site with a third exporter and consistent labels: the SYN flood is detected, whereas much of the benign traffic

TABLE 14. Per-category detection, mean over 20 split seeds at $\tau = 0.5$. “Ben.” is specificity; other columns are recall. BF: brute force. Categories absent from $\mathcal{D}_B$ can be tested only in the reverse direction.

Model	$\mathcal{D}_A \rightarrow \mathcal{D}_B$			$\mathcal{D}_B \rightarrow \mathcal{D}_A$ (absent from $\mathcal{D}_B$: DDoS, BF, web)					
	Ben.	DoS	Probe	Ben.	DoS	Probe	DDoS	BF	Web
LR	0.98	0.25	0.64	0.71	0.85	0.92	0.96	0.57	0.30
DT	0.44	0.65	0.29	0.82	0.44	0.47	0.48	0.43	0.13
RF	0.46	0.74	0.19	0.89	0.41	0.55	0.49	0.34	0.00
XGB	0.57	0.58	0.17	0.89	0.73	0.88	0.85	0.35	0.01
HGB	0.54	0.63	0.18	0.87	0.72	0.87	0.83	0.34	0.01
MLP	0.46	0.68	0.25	0.86	0.71	0.73	0.81	0.36	0.00

TABLE 15. Transfer from $\mathcal{D}_A$ to $\mathcal{D}_B$ under alternative feature sets, adaptation, and model families, mean over 20 split seeds. CORAL reads unlabeled target features.

Arm	Model	Src. BA	Tgt. BA	Tgt. AUC	Tgt. FPR
Shared space (18)	LR	0.837	0.627	0.701	0.023
	DT	0.822	0.533	0.526	0.558
	RF	0.808	0.581	0.578	0.541
	XGB	0.864	0.562	0.572	0.429
	HGB	0.867	0.566	0.581	0.463
Exporter-robust subset (9)	MLP	0.912	0.553	0.557	0.539
	LR	0.804	0.510	0.583	0.854
	DT	0.883	0.480	0.488	0.177
	RF	0.894	0.507	0.469	0.206
	XGB	0.925	0.511	0.533	0.089
CORAL (reads target features)	HGB	0.939	0.521	0.541	0.090
	MLP	0.928	0.534	0.499	0.039
	LR	0.836	0.503	0.646	0.902
	DT	0.812	0.500	0.504	0.870
	RF	0.795	0.481	0.515	0.992
Benign-only novelty	XGB	0.906	0.473	0.478	0.906
	HGB	0.875	0.481	0.484	0.915
	MLP	0.930	0.602	0.632	0.552
	IF	0.503	0.492	0.464	0.024
	AE	0.455	0.479	0.484	0.074

TABLE 16. In-distribution cost of the shared space on $\mathcal{D}_A$ flows: all 46 non-port Zeek features against the 18 shared columns, same splits. Loss interval: Nadeau–Bengio corrected, 95%.

Model	BA full	BA shared	Loss [95% CI]	F_1 full	F_1 shared
LR	0.918	0.837	+0.082 [-0.04, +0.20]	0.725	0.626
DT	0.861	0.822	+0.039 [-0.12, +0.20]	0.663	0.619
RF	0.846	0.808	+0.038 [-0.03, +0.11]	0.726	0.701
XGB	0.885	0.864	+0.020 [-0.05, +0.09]	0.677	0.646
HGB	0.893	0.867	+0.026 [-0.11, +0.16]	0.680	0.647
MLP	0.917	0.912	+0.005 [-0.08, +0.10]	0.708	0.655

of the site is not recognized as benign.

G. ALERT BURDEN

Table 18 reports pooled operating points at $\tau = 0.5$ for three evaluation sets, together with 95% cluster-bootstrap intervals. On the radio layer, the six detectors have a pooled window-level false positive rate of 0.22 to 0.33, which corresponds to 50 to 74 false alerts per benign UE-hour, and, at $\pi = 0.002$, an operational precision of 0.0055 to 0.0080, compared with a corpus precision of 0.90 to 0.93. These point estimates are not stable. The false positive rate depends on which of the ten benign sessions are included in the test set (Table 7), and the bootstrap interval of the pooled rate spans 0.01 to 0.65

across architectures, or 3 to 146 false alerts per benign UE-hour. Likewise, the interval for operational precision at $\tau = 0.5$ spans 0.0028 to 0.13. Thirty sessions are insufficient to determine the alert burden of the radio layer, and they show only that it depends on the benign conditions that a detector encounters. On held-out $\mathcal{D}_A$ flows, the false positive rate is 0.066 to 0.201, which at $\lambda_b = 240,000$ benign flows per hour corresponds to 15,726 to 48,289 false alerts per hour. On $\mathcal{D}_B$, the false positive rate is 0.023 to 0.558, or 5,620 to 133,990 false alerts per hour. Without the benign copies described in Section VI-C, it is 0.026 to 0.183, which still amounts to 6,158 to 44,003 false alerts per hour, and the operational precision is 0.0072 to 0.0398.

Increasing the decision threshold improves precision only on held-out $\mathcal{D}_A$ flows (Fig. 8). Because the best threshold is selected on the evaluation data, the following values should be regarded as upper bounds. With recall kept above 10%, high thresholds reach an operational precision of 0.42 on held-out $\mathcal{D}_A$ flows, whereas the best value achieved by any detector on $\mathcal{D}_B$ is 0.026, or 0.063 without the copies. On the radio layer, the best pooled value is 0.036, but its bootstrap interval reaches 1.0 for 4 of the six architectures. This occurs because a resample that omits the benign sessions that are flagged almost entirely leaves thresholds at which no false positive occurs. The relation between corpus precision and operational precision is determined arithmetically by π . Varying π from 10^{-4} to 0.5 changes the ratio between the two from 2,627 to 1.18 (Fig. 9), so an operational precision is meaningful only when the assumed prevalence is stated. *Estimation of operational precision.* PPV is strongly nonlinear in the false positive rate near zero, so a fold that happens to contain no false positive yields $PPV = 1$ regardless of its recall. Averaging PPV over folds consequently favors the architecture that happens to draw the largest number of such folds. On the radio layer, the same counts yield a ratio between the best and the worst architecture of 1.45 when pooled, 1.89 as a median over folds, and 16.0 as a mean over folds (Table 19). Pooled counts are therefore used throughout this paper.

H. CALIBRATION AND PRIOR SHIFT

Under transfer, the scores are poorly calibrated (Fig. 10). We fitted three calibrators on a group-disjoint calibration fold drawn from the source (Table 20). Whether a calibrator can alter the achievable operating points depends on whether it

TABLE 17. Transfer to the third corpus $\mathcal{D}_C$ (5G core testbed, NFStream flows [22]): balanced accuracy on held-out source data and on $\mathcal{D}_C$, target ROC-AUC, recall per attack type, and the share of the core's background signalling flagged (Bkg.). Mean over 20 split seeds from $\mathcal{D}_A$ and 10 from $\mathcal{D}_B$, at $\tau = 0.5$.

Model	Src. BA	Tgt. BA	Trained on $\mathcal{D}_A$					Trained on $\mathcal{D}_B$	
			AUC	SYN	ICMP	PFCP	Bkg.	Tgt. BA	AUC
LR	0.837	0.742	0.800	1.00	0.10	0.00	0.27	0.517	0.726
DT	0.822	0.658	0.696	1.00	1.00	0.79	0.67	0.422	0.384
RF	0.808	0.596	0.926	1.00	1.00	0.99	0.81	0.540	0.697
XGB	0.864	0.640	0.888	1.00	1.00	0.74	0.70	0.688	0.814
HGB	0.867	0.698	0.932	1.00	1.00	0.80	0.61	0.565	0.632
MLP	0.912	0.691	0.837	1.00	0.20	0.61	0.44	0.848	0.766
Maj.	0.500	0.500	0.500	1.00	1.00	1.00	1.00	0.500	0.500

TABLE 18. Pooled operating points at $\tau = 0.5$, 20 split seeds. FPR interval: 95% cluster bootstrap resampling split seeds and clusters (radio sessions, $\mathcal{D}_A$ source addresses, $\mathcal{D}_B$ capture files; for flows from a deterministic re-run on 10 seeds, EXP-056, on which the “w/o conflicts” block is also computed). “w/o conflicts”: $\mathcal{D}_B$ without the benign flows whose record also occurs with an attack label. PPV at $\pi = 0.002$. [†] False alerts per benign UE-hour (radio, 225 windows per UE-hour) or per hour at $\lambda_b = 240,000$ benign flows (flows). [‡] Highest pooled PPV over all thresholds with recall at least 0.10.

Corpus	Model	TPR	FPR [95% CI]	Corpus P	PPV	False alerts [†]	Best PPV [‡]
$\mathcal{D}_A$ radio	LR	0.798	0.223 [0.017, 0.547]	0.921	0.0071	50	0.0130
	DT	0.910	0.329 [0.086, 0.639]	0.900	0.0055	74	0.0055
	RF	0.954	0.267 [0.024, 0.650]	0.921	0.0071	60	0.0360
	XGB	0.931	0.251 [0.013, 0.606]	0.924	0.0074	57	0.0116
	HGB	0.945	0.262 [0.018, 0.615]	0.922	0.0072	59	0.0086
	MLP	0.895	0.222 [0.019, 0.560]	0.929	0.0080	50	0.0109
	Maj.	1.000	1.000 [1.000, 1.000]	0.766	0.0020	225	0.0020
$\mathcal{D}_A$ network	LR	0.892	0.201 [0.126, 0.445]	0.992	0.0088	48 289	0.0297
	DT	0.852	0.157 [0.068, 0.498]	0.994	0.0108	37 658	0.0120
	RF	0.920	0.198 [0.121, 0.611]	0.993	0.0092	47 568	0.4172
	XGB	0.884	0.090 [0.027, 0.373]	0.997	0.0193	21 625	0.2735
	HGB	0.881	0.066 [0.031, 0.112]	0.998	0.0262	15 726	0.2727
	MLP	0.905	0.071 [0.043, 0.349]	0.997	0.0250	16 996	0.2189
	Maj.	1.000	1.000 [1.000, 1.000]	0.967	0.0020	240 000	0.0020
$\mathcal{D}_B$ (transfer)	LR	0.277	0.023 [0.009, 0.079]	0.948	0.0232	5 620	0.0258
	DT	0.624	0.558 [0.111, 0.783]	0.633	0.0022	133 990	0.0022
	RF	0.703	0.541 [0.121, 0.796]	0.667	0.0026	129 843	0.0027
	XGB	0.552	0.429 [0.013, 0.687]	0.666	0.0026	102 890	0.0026
	HGB	0.596	0.463 [0.031, 0.699]	0.665	0.0026	111 104	0.0026
	MLP	0.646	0.539 [0.030, 0.745]	0.649	0.0024	129 320	0.0026
	Maj.	1.000	1.000 [1.000, 1.000]	0.607	0.0020	240 000	0.0020
$\mathcal{D}_B$ w/o conflicts	LR	0.269	0.057 [0.038, 0.086]	0.947	0.0094	13 629	0.0102
	DT	0.662	0.183 [0.103, 0.294]	0.931	0.0072	44 003	0.0077
	RF	0.719	0.165 [0.118, 0.258]	0.943	0.0087	39 561	0.0629
	XGB	0.531	0.026 [0.012, 0.047]	0.987	0.0398	6 158	0.0631
	HGB	0.548	0.047 [0.030, 0.080]	0.978	0.0227	11 315	0.0587
	MLP	0.612	0.053 [0.026, 0.102]	0.977	0.0225	12 794	0.0306
	Maj.	1.000	1.000 [1.000, 1.000]	0.790	0.0020	240 000	0.0020

TABLE 19. Operational precision on the radio layer at $\tau = 0.5$, $\pi = 0.002$, estimated three ways from the same 20 folds.

Model	Folds with FP = 0	Pooled	Median	Mean
LR	1	0.0071	0.0118	0.0815
DT	0	0.0055	0.0070	0.0137
RF	1	0.0071	0.0092	0.0814
XGB	4	0.0074	0.0095	0.2185
HGB	2	0.0072	0.0091	0.1291
MLP	0	0.0080	0.0132	0.0476

preserves the order of the scores. Temperature scaling preserves this order and changes ROC-AUC by at most $2e-05$. Platt scaling is monotone, but its fitted slope can be negative, and for DT it inverted the ranking in 1 of 10 seeds, changing

ROC-AUC by 0.95. In that seed, the scores of the tree carried no ranking information on the calibration fold (ROC-AUC 0.48 on 96 held-out source addresses), although they ranked the test fold well (0.97). Platt scaling fitted the calibration fold correctly with a slope of -0.55 and, as a result, inverted the ranking on the test fold. This failure is attributable to small group-disjoint calibration folds, which need not resemble the test fold, rather than to Platt scaling itself. Isotonic regression is monotone but piecewise constant, so it merges scores into ties and, on its fitting data, yields the ROC convex hull [52]. Its largest change in ROC-AUC ranged from 0.04 to 0.21 across architectures. None of the calibrators raises the target macro- F_1 above that of the raw scores for any architecture. Because they are fitted on a source containing 94.63% attacks,

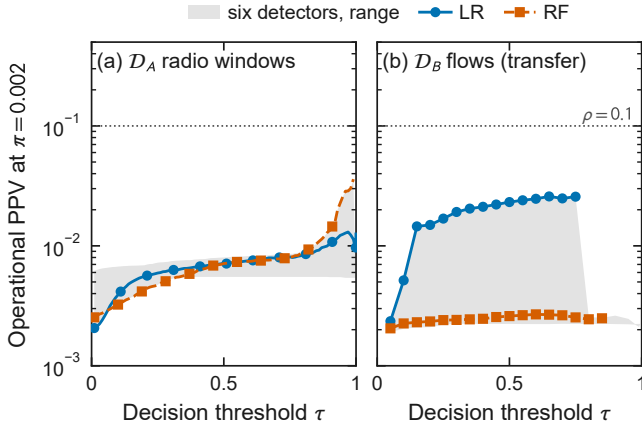


FIGURE 8. Pooled operational precision at $\pi = 0.002$ against the decision threshold on (a) the $\mathcal{D}_A$ radio layer and (b) $\mathcal{D}_B$ flows after transfer, at thresholds that keep recall at or above 0.10. The gray band spans the six architectures, and the dotted line marks the precision bound $\rho = 0.1$ of the deployability test, which is not reached at any threshold on either evaluation set.

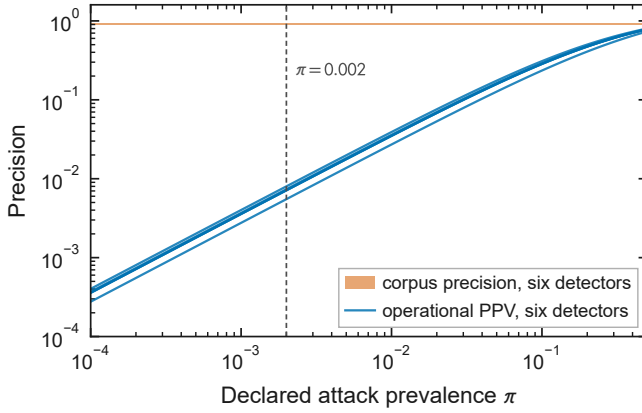


FIGURE 9. Pooled operational precision at $\tau = 0.5$ against the declared attack prevalence π for the six radio-layer detectors (one line each), with their corpus precision as a band. The dashed rule marks $\pi = 0.002$. Operational precision is computed from the same confusion counts with Eq. (4), and only the declared prevalence is varied.

they shift the scores toward that prior and flag more benign traffic on $\mathcal{D}_B$.

A standard remedy for prior shift is to rescale the scores to the target prior [48], which is, however, unknown in a deployment. Two estimators recover this prior from unlabeled data under the assumption that only the class balance changes, namely expectation maximization [48] and black-box shift estimation [49]. In the present setting, this assumption does not hold. For a true target prior of 0.607, the mean estimates per architecture range from 0.16 to 0.86, and 41% of the individual estimates fall below 0.05 or above 0.95. Moreover, rescaling with the true prior, which is an oracle setting, does not outperform the raw scores (Table 20). The limiting factor is therefore the discriminability of the detectors on the target, with ROC-AUC of 0.53 to 0.70, rather than their calibration.

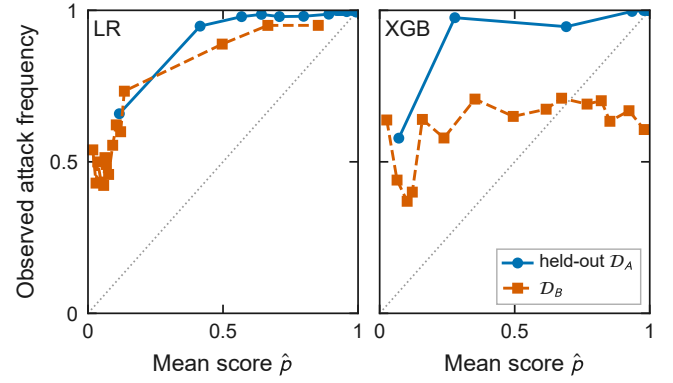


FIGURE 10. Reliability diagrams for LR and XGB on held-out $\mathcal{D}_A$ and on $\mathcal{D}_B$, fifteen equal-mass bins pooled over 20 split seeds.

I. DECISION LATENCY

Table 21 reports the per-call latency on one host for three implementations of each model. The first calls scikit-learn with a one-row pandas DataFrame, which corresponds to how the model is invoked in the experiment code. The second calls scikit-learn with a NumPy array on a single thread. The third uses ONNX Runtime on a single thread and is applied only where the exported model reproduces the decisions of scikit-learn (12 of 12 exports, with a decision agreement of at least 1.0000). A no-op call has a 99th percentile of 0.001 ms, so the overhead of the platform does not affect the comparisons.

The implementation has a larger effect on latency than the architecture (Fig. 11). For LR, the DataFrame path requires 0.44 ms at the median, compared with 0.11 ms for an array, and the difference is due to input validation. For RF, predicting a single row through the default thread pool requires 29 ms at the median, compared with 5.6 ms on one thread and 0.012 ms in ONNX Runtime. Across architectures, the 99th percentile for ONNX Runtime lies between 0.014 and 0.035 ms on the radio layer, whereas it reaches up to 21.5 ms for scikit-learn with arrays. A ranking of architectures by latency is largely a ranking of implementations, and a comparison with measurements obtained on other hosts or in other programming languages [21] provides no information about the architectures themselves. For context only, Obiuevwi et al. report inference times in the microsecond range for logistic regression and a small MLP inside an OpenAirInterface and FlexRIC near-real-time RIC [21], which is within an order of magnitude of our ONNX Runtime measurements.

For a window-level detector, the decision path consists of window aggregation followed by inference. Aggregating one 16-record window requires 0.064 ms at the 99th percentile, and the upper bound of the 99th percentile of aggregation plus ONNX inference is at most 0.12 ms across the architectures with a verified export. On this host, the path would meet any budget in the specified range. Whether it would do so in a deployment depends on the terms that were not measured, namely indication decoding, control encoding, and queueing inside a RIC. A repetition of the radio measurement on the same host, in which every call was recorded for Fig. 11,

TABLE 20. Calibration over 10 split seeds. Left: largest absolute change in ROC-AUC caused by each calibrator, over seeds and both evaluation sets. Right: target macro- F_1 at $\tau = 0.5$ for raw scores, Platt and isotonic calibration, and Platt followed by prior correction with the EM estimate or the true (oracle) target prior.

Model	max $ \Delta \text{ROC-AUC} $			Target macro- F_1 at $\tau = 0.5$				
	Temp.	Platt	Iso.	Raw	Platt	Iso.	+EM	+Oracle
LR	$< 10^{-6}$	1×10^{-6}	0.209	0.538	0.415	0.391	0.327	0.478
DT	2×10^{-5}	0.946	0.170	0.538	0.395	0.405	0.447	0.436
RF	$< 10^{-6}$	$< 10^{-6}$	0.050	0.579	0.504	0.470	0.440	0.488
XGB	$< 10^{-6}$	$< 10^{-6}$	0.150	0.555	0.450	0.394	0.456	0.518
HGB	$< 10^{-6}$	$< 10^{-6}$	0.095	0.549	0.429	0.399	0.479	0.507
MLP	2×10^{-5}	2×10^{-5}	0.045	0.578	0.405	0.408	0.475	0.534

yielded medians 1.02 to 2.1 times those in Table 21 (median ratio 1.46). Per-call latency on a host without CPU isolation thus depends on the state of the host as well as on the model. Nevertheless, the upper 95% bound of the 99th percentile of aggregation plus ONNX inference remained at most 0.36 ms in the repeated measurement.

For a flow-level detector, feature construction is more costly than inference. On the three DDoS captures of $\mathcal{D}_A$, the vectorized exporter produces flow records at 0.20 to 0.90 ms per flow, which is 22.8 to 41.6 times faster than the per-packet reference implementation on the same captures (4.49 to 37.34 ms per flow), and the cost per flow increases with the number of packets per flow. On the 2 captures that were compared record by record, the two exporters agree exactly in packets and bytes on all 38,032 shared IPv4 flow keys. The reference implementation additionally emits 22 and 41 IPv6 link-local keys, which the vectorized exporter does not parse because it supports IPv4 only. Mapping one exported record onto the shared space with the pandas code of our pipeline requires an additional 8.5 ms at the median, whereas ONNX inference on flow features requires at most 0.037 ms at the 99th percentile. These figures describe a Python pipeline on a single host. They indicate that the cost lies mainly in feature construction, but they do not indicate what a native exporter at the user plane would cost.

VII. DISCUSSION

This section discusses the implications of the results for O-RAN practice, applies the deployability test, and relates the results to previously reported findings.

A. IMPLICATIONS FOR O-RAN PRACTICE

Data path. E2SM-KPM delivers aggregated measurements for each reporting period and does not carry packets. A window-level detector can operate inside the RIC on KPM reports, whereas a flow-level detector requires a flow exporter located near the user plane, either in the central unit or at the user-plane function, as well as a means of transporting its records to the xApp, for example through a custom E2 service model or the non-real-time data collection path. This transport contributes to t_{ind} in Eq. (5), which was not measured in this study.

Detection delay. Computational latency accounts for only a small part of the total time to detection. A window-level

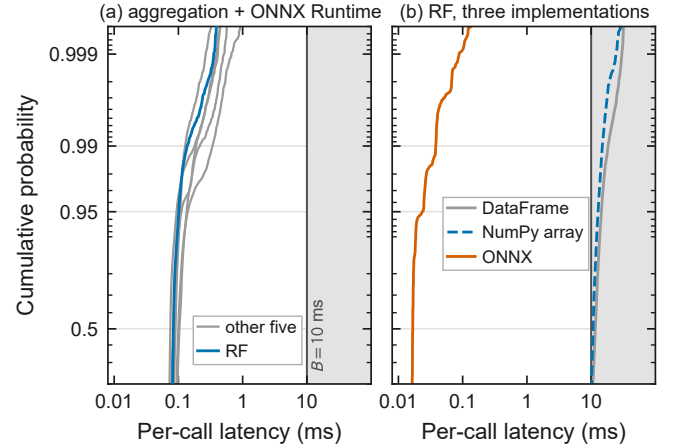


FIGURE 11. Empirical distribution of per-call latency for the radio layer on one host, 20,000 calls per stage, on a logit scale so that the median, 95th, and 99th percentiles are all visible. (a) The decision path of the deployability test, window aggregation followed by ONNX Runtime, for the six architectures. (b) One architecture, RF, under the three implementations of Table 21. The shaded region lies beyond $B = 10$ ms, the smallest budget in the specified range. Only feature construction and inference are timed; indication decoding, control encoding, and queueing, which need a RIC in the path, are not. The curves come from a repeat of the measurement that kept every call (EXP-060); Table 21 reports the original run.

decision cannot be made before the end of its window (16 s in this study), and a flow-level decision cannot be made before the flow record is exported, which, for a long connection, requires an idle or active timeout to expire. The KPM reporting period and the flow timeouts should be reported together with the computational latency.

Acting on alerts. An alert that triggers an E2SM-RC action, such as isolating or steering a UE, acts on a false alert in the same way as on a true one. At the false positive rates reported in Table 18, automatic mitigation on the radio layer would act against each benign UE 50 to 74 times per hour. Alerts of this quality may inform an operator or a non-real-time policy through A1, but they should not trigger automatic control actions.

Admission and monitoring. The deployability test of Eq. (6) can serve as an admission check before an xApp is enabled at a new site, provided that labeled benign and attack traffic is available from that site. After admission, unsupervised shift detection [53] can indicate when the incoming traffic departs

TABLE 21. Per-call latency in milliseconds on one host (Section V): 20,000 calls after 1,000 warm-up calls, one thread per library, garbage collection enabled. Emulated: no RIC in the path. Radio path $\bar{p}_{99}$: upper 95% bound of the 99th percentile of window aggregation plus ONNX inference. Dashes: no export that reproduced scikit-learn's decisions.

Model	sklearn, DataFrame		sklearn, array		ONNX Runtime		Radio path	Flow layer, p_{99}	
	p_{50}	p_{99}	p_{50}	p_{99}	p_{50}	p_{99}	$\bar{p}_{99}$	array	ONNX
LR	0.44	0.72	0.11	0.17	0.010	0.023	0.12	0.16	0.025
DT	0.31	0.52	0.03	0.04	0.010	0.014	0.11	0.06	0.024
RF	6.25	12.99	5.57	21.50	0.012	0.024	0.10	4.45	0.029
XGB	0.73	1.00	0.16	0.32	0.011	0.021	0.09	0.40	0.024
HGB	1.48	2.72	1.17	1.92	0.013	0.035	0.12	1.22	0.034
MLP	0.43	0.67	0.11	0.21	0.017	0.035	0.11	0.15	0.037

from the training distribution. Because the corpora studied here can be distinguished with a balanced accuracy of 0.999, such a monitor would raise an alarm immediately.

B. APPLYING THE DEPLOYABILITY TEST

We instantiate Eq. (6) with the bounds described below. These bounds were chosen by us, since no O-RAN specification defines them, and an operator would set them according to its own traffic and staffing. The generalization bound is $\delta = 0.10$ in balanced accuracy and is applied to the upper end of the corrected 95% interval. The alert term requires a threshold that detects at least half of the attacks ($r_{\min} = 0.5$), for which at least one alert in ten is a true alert ($\rho = 0.1$ at $\pi = 0.002$), and which raises at most $A_{\max} = 200$ false alerts per hour at $\lambda_b = 240,000$ benign flows per hour. All alert conditions are evaluated on $\mathcal{D}_B$, because a detector is deployed on the target rather than on the source. The latency term uses the strictest budget permitted by the specification, $B = 10$ ms. Because the upper bound of the 99th percentile of the radio decision path is below 1 ms for every architecture (Section VI-I), this term is satisfied for any budget in the specified range, subject to the caveat that indication decoding, control encoding, and queueing were not measured.

Table 22 presents the result, according to which none of the architectures passes the test (0 of six). Every architecture fails the generalization term, with upper bounds of 0.32 to 0.50, and every architecture also fails the alert term by a wide margin. At a recall of at least 0.5, the highest operational precision on $\mathcal{D}_B$ is 0.0027, and the smallest number of false alerts per hour is 95,507. The verdict is therefore robust to the choice of bounds. It holds for any ρ above 0.0027, any $A_{\max}$ below 95,507 per hour, and any δ below 0.32, and it does not depend on the latency term. The verdict is also unaffected by the label conflicts described in Section VI-C. When the test is evaluated on $\mathcal{D}_B$ without the benign copies, 0 of six architectures pass, the highest operational precision at a recall of at least 0.5 is 0.0419, and the smallest number of false alerts per hour is 5,603. Fig. 12 shows how far the bounds would have to be relaxed. An architecture satisfies the alert term only if an operator accepts several thousand false alerts per hour and a precision of a few percent, and it satisfies the generalization term only if δ exceeds 0.32. The latency term would become relevant only if the first two terms were satisfied, and, on the basis of our off-platform measurements,

TABLE 22. Deployability test of Eq. (6) per architecture. $\bar{\Delta}_{BA}$: upper 95% bound of the balanced-accuracy loss (pass if ≤ 0.10). Best PPV and minimum false alerts per hour over thresholds with recall ≥ 0.5 on $\mathcal{D}_B$ (pass if PPV ≥ 0.1 and alerts ≤ 200). $B_{\min}$: smallest budget met by the upper bound of the radio decision path's 99th percentile, timed off-platform.

Model	$\bar{\Delta}_{BA}$	Gen.	Best PPV	Min. alerts/h	Alert	$B_{\min}$ (ms)	$\mathcal{P}$
LR	0.32	×	0.0024	166 328	×	1	0
DT	0.50	×	0.0022	118 342	×	1	0
RF	0.47	×	0.0027	98 410	×	1	0
XGB	0.47	×	0.0026	95 507	×	1	0
HGB	0.47	×	0.0026	99 277	×	1	0
MLP	0.50	×	0.0026	105 744	×	1	0

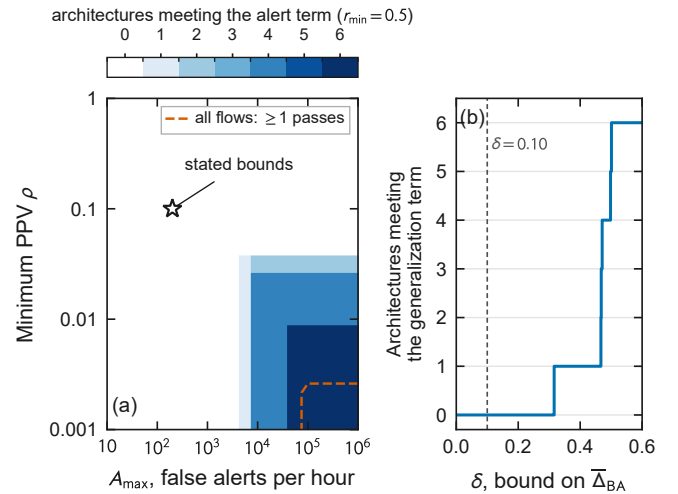


FIGURE 12. Sensitivity of the deployability test to its bounds. (a) Number of the six architectures meeting the alert term ($r_{\min} = 0.5$, $\pi = 0.002$, $\lambda_b = 240,000$ flows per hour) on $\mathcal{D}_B$ without the benign copies, as the alert bound $A_{\max}$ and the precision bound ρ vary; on all flows, at least one architecture passes only below and to the right of the dashed line. The star marks the bounds we use. (b) Number of architectures meeting the generalization term as δ varies. A detector passes the test only if both terms are satisfied.

it does not determine the verdict.

Attainability of the test. A test that no detector passes might simply be too strict. The published 5G-NIDD pipeline described in Section VI-C indicates that this is not the case. On its own random split, with the alert term evaluated on that split, 2 of its five classifiers pass the test, with an operational precision of up to 0.79 at $\pi = 0.002$ and as few as 128 false alerts per hour at $\lambda_b = 240,000$. The same pipeline

fails the test at the next rung, when the two position counters are removed, and at every subsequent rung, both on all flows and on the flows without the benign copies (Table 11). The informativeness of the test is therefore limited by the evaluation to which it is applied. When applied to a random split that still contains record positions, it merely confirms the result of that split, and the pass disappears as soon as the evaluation moves one step closer to deployment conditions. The detectors examined in this paper fail the test under the evaluations closest to deployment that the available corpora permit.

C. RELATION TO REPORTED RESULTS

Most published results for O-RAN and 5G intrusion detection are in-distribution scores obtained on corpora that differ from ours, and our own in-distribution scores under a random split are similarly high (Table 5). A direct numerical comparison would compare evaluation protocols rather than detectors. The results presented above instead support a statement about which of the commonly reported quantities are predictive of deployment. Held-out macro- F_1 under a random split is not, whereas the quantities listed in the reporting requirements of Section X are those that altered the conclusions of this study.

VIII. REPRODUCIBILITY AND ETHICS

The code, the configuration, the seeded split procedures from which every split can be regenerated deterministically, the raw per-seed results, and the generators for every table, figure, and in-text number are available at <https://github.com/adeliusa486/oran-ids-deployability>. A single script regenerates every number, table, and figure of this paper from the provided per-seed results and verifies them against the manuscript. The same script can also re-run each experiment from the raw corpora, whose files it verifies by SHA-256, and the file `REPRODUCE.md` in the repository describes the procedure. All three corpora are publicly available: $\mathcal{D}_A$ is distributed by its authors under CC-BY-4.0 [39], $\mathcal{D}_B$ is distributed by its authors [37], and $\mathcal{D}_C$ is available from the repository of its authors [22]. Because the latter repository states no license, we do not redistribute $\mathcal{D}_C$. The traffic originates from testbed devices and a test network and not from subscribers, so no personal data was processed. Reporting the conditions under which detectors fail carries a modest dual-use risk, which we consider to be outweighed by the benefit to operators of knowing that reported accuracy does not necessarily transfer to their networks.

IX. LIMITATIONS

Flow exporters. $\mathcal{D}_A$, $\mathcal{D}_B$, and $\mathcal{D}_C$ were produced by three different tools (Zeek, Argus, and NFStream), and it was not possible to re-extract $\mathcal{D}_A$ or $\mathcal{D}_B$ with a common exporter, because our copy of $\mathcal{D}_B$ contains Argus flow records only and no Zeek installation was available on the measurement host. The domain classifier and the robust subset show that the corpora differ in traffic composition as well as in flow segmentation, but they do not quantify how much of the

transfer gap a common exporter would remove. The base-station split within $\mathcal{D}_B$ (Section VI-C) changes the site while keeping the exporter fixed, and it therefore addresses part of this question.

Target labels. Of the benign flows in $\mathcal{D}_B$, 59% are exact copies of records that are labeled as UDP flood elsewhere in the corpus (Section VI-C). Because the published files do not indicate which label is correct, every operating point on $\mathcal{D}_B$ is reported both with and without these flows. In addition, the assignment of the two capture passes to the two base stations is inferred from the repeated capture schedule and the restarted row index, since it is not stated in the dataset documentation.

Radio capture schedule. Nine of the ten benign radio sessions were recorded six weeks before the first attack session, mostly with a single UE, and every radio label is a session label. A radio detector may learn to separate capture periods in addition to attacks. The only benign session recorded during the attack period is flagged almost entirely, but the same holds for two early sessions (Table 7), and ten benign sessions are not sufficient to distinguish between the two explanations.

Number of corpora. Three corpora are sufficient to establish that the transfer gap recurs, but not to characterize how it varies across deployments. The third corpus originates from a 5G core testbed rather than from an O-RAN deployment, contains only flood attacks and a control-plane attack, and comprises three capture files, which are too few for cluster-based intervals. No further O-RAN corpus with paired radio telemetry was available to us.

Latency. All latency figures were obtained by emulation on a single Windows host without CPU isolation, and t_{ind} , t_{act} , and t_q were not measured. CPU and memory utilization under load were not measured either.

Operational parameters and the deployability test. The prevalence $\pi = 0.002$ and the benign arrival rate λ_b are declared values and were not measured on an operational network. Both are varied over a range, and operational precision depends on π alone. The bounds of the deployability test were chosen by us (Section VII-B). Although Fig. 12 reports the verdict under other bounds, an operator may weigh recall, precision, and alert volume differently, and the test does not account for detection delay or for the cost of a missed attack.

Sample size. The radio layer comprises 30 capture sessions, each of which contains a single scenario. Session and attack scenario are therefore confounded, and the protocol and temporal analyses can detect only large effects. Because only ten benign sessions are available, the false positive rate of the radio layer, and hence its operational precision, carries a wide interval (Section VI-G). The transfer statistics vary the training set over 20 seeds but evaluate a single fixed target corpus, so they do not describe the variability across other deployments.

Scope. This study evaluates binary detection only. Multi-class attribution, feature attribution [62], and the fusion of radio and flow features are left for future work. The published summary files cannot be joined record by record, so fusion

requires the raw archives, as in [31]. Furthermore, three attack categories of $\mathcal{D}_A$ are absent from $\mathcal{D}_B$ and can therefore be tested in one direction only.

X. CONCLUSION AND REPORTING REQUIREMENTS

This study shows that a held-out score on a single corpus provides limited information about the behavior of a detection xApp in a different deployment. On the corpora examined here, a random split inflates the score, and a model-free lookup achieves a comparable gain, which is consistent with the recognition of capture sessions. The capture schedule of the radio corpus does not permit an estimate of temporal drift, and the false positive rate depends largely on the benign session that a detector encounters. A published accuracy of 99.9% on the target corpus depends on record-position fields, and half of that corpus carries labels that cannot be determined from the record content. On flows with consistent labels, detectors transferred from the source corpus achieve a balanced accuracy of 0.61 to 0.78, which is below the level attainable within the target corpus. On a third corpus, the detectors identify the SYN flood but flag a large share of the benign traffic. At a realistic base rate, no threshold yields a usable operational precision, and latency, the only criterion that is satisfied, was measured off-platform. We therefore propose that studies claiming a deployable O-RAN detection capability report at least the following: (i) performance on a corpus collected independently of the training data, in a prevalence-invariant metric such as balanced accuracy or ROC-AUC, beside input-blind baselines and a reference trained inside that corpus; (ii) the split protocol and its group key, results under a group-disjoint split, and the category coverage of any time-based split; (iii) the false positive rate on benign traffic, with its unit, the resulting alert rate, and the operational precision at a stated base rate, all from pooled confusion counts with intervals that resample clusters; (iv) the operating threshold, how it was chosen, and the recall it gives up; (v) intervals that account for overlap between resampled splits; (vi) tail latency with its stages, implementation, and host, measured on a real RIC before any conformance claim is made; and (vii) evidence that the labels are a function of the features used, with no identical records under opposite labels, and that no record-position or identifier field is a feature.

REFERENCES

- [1] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1376–1411, 2023. doi: 10.1109/COMST.2023.3239220
- [2] A. S. Abdalla, P. S. Upadhyaya, V. K. Shah, and V. Marojevic, "Toward next generation open radio access networks: What O-RAN can and cannot do!" *IEEE Network*, vol. 36, no. 6, pp. 206–213, 2022. doi: 10.1109/MNET.108.2100659
- [3] S. Niknam, A. Roy, H. S. Dhillon, S. Singh, R. Banerji, J. H. Reed, N. Saxena, and S. Yoon, "Intelligent O-RAN for beyond 5G and 6G wireless networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 215–220. doi: 10.1109/GCWkshps56602.2022.10008676
- [4] L. Bonati, S. D'Oro, M. Polese, S. Basagni, and T. Melodia, "Intelligence and learning in O-RAN for data-driven NextG cellular networks," *IEEE Communications Magazine*, vol. 59, no. 10, pp. 21–27, 2021. doi: 10.1109/MCOM.101.2001120
- [5] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis, D. Kumar, C. Lever, Z. Ma, J. Mason, D. Menscher, C. Seaman, N. Sullivan, K. Thomas, and Y. Zhou, "Understanding the Mirai botnet," in *Proc. 26th USENIX Security Symposium*, 2017, pp. 1093–1110. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis>
- [6] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and other botnets," *Computer*, vol. 50, no. 7, pp. 80–84, 2017. doi: 10.1109/MC.2017.201
- [7] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "CoLo-RAN: Developing machine learning-based xApps for open RAN closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5787–5800, 2023. doi: 10.1109/TMC.2022.3188013
- [8] R. Schmidt, M. Irazabal, and N. Nikaein, "FlexRIC: An SDK for next-generation SD-RANs," in *Proc. ACM CoNEXT*, 2021, pp. 411–425. doi: 10.1145/3485983.3494870
- [9] H. Wen, P. Porras, V. Yegneswaran, A. Gehani, and Z. Lin, "5G-Spector: An O-RAN compliant layer-3 cellular attack detection service," in *Proc. Network and Distributed System Security Symposium (NDSS)*, 2024. doi: 10.14722/ndss.2024.24527
- [10] A. Scalingi, S. D'Oro, F. Restuccia, T. Melodia, and D. Giustiniano, "Det-RAN: Data-driven cross-layer real-time attack detection in 5G open RANs," in *Proc. IEEE Conf. on Computer Communications (INFOCOM)*, 2024, pp. 41–50. doi: 10.1109/INFOCOM52122.2024.10621223
- [11] S. Samarakoon, Y. Siriwardhana, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Ylianttila, "5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network," arXiv:2212.01298, 2022. [Online]. Available: <https://arxiv.org/abs/2212.01298>
- [12] L. Ilias, G. Doukas, V. Lamprou, C. Ntanos, and D. Askounis, "Convolutional neural networks and mixture of experts for intrusion detection in 5G networks and beyond," arXiv:2412.03483, 2024, the weighted F_1 cited is from the abstract of v2 (May 2025); a revised version appeared in *Front. Artif. Intell.*, vol. 8, art. 1708953, 2026, doi:10.3389/frai.2025.1708953. [Online]. Available: <https://arxiv.org/abs/2412.03483>
- [13] S. Layeghy and M. Portmann, "Explainable cross-domain evaluation of ML-based network intrusion detection systems," *Computers and Electrical Engineering*, vol. 108, p. 108692, 2023. doi: 10.1016/j.compeleceng.2023.108692
- [14] S. Layeghy, M. Gallagher, and M. Portmann, "Benchmarking the benchmark — comparing synthetic and real-world network IDS datasets," *Journal of Information Security and Applications*, vol. 80, p. 103689, 2024. doi: 10.1016/j.jisa.2023.103689
- [15] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security*, vol. 3, no. 3, pp. 186–205, 2000. doi: 10.1145/357830.357849
- [16] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. IEEE Symposium on Security and Privacy*, 2010, pp. 305–316. doi: 10.1109/SP.2010.25
- [17] ETSI, "Publicly available specification (PAS); E2 interface: General aspects and principles (O-RAN.WG3.E2GAP-R003-v04.01)," European Telecommunications Standards Institute, ETSI TS 104 038 V4.1.0, Oct. 2024. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/104000_104099/104038/04.01.00_60/ts_104038v040100p.pdf
- [18] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *Proc. 31st USENIX Security Symposium*, 2022, pp. 3971–3988. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity22/presentation/arp>
- [19] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in *Proc. 28th USENIX Security Symposium*, 2019, pp. 729–746.
- [20] A. Abrahameem and A. Edhirig, "Bidirectional cross-dataset generalization and label-efficient adaptation for 5G network intrusion detection," *Wadi Alshatti University Journal of Pure and Applied Sciences*, vol. 4, no. 2, Aug. 2026. [Online]. Available: <https://waujpas.com/index.php/journal/article/view/509>
- [21] L. Obiuewewi, K. J. Rechowicz, S. Jayarathna, S. H. Bouk, F. Afrin, C. N. Barati, N. Moghim, V. Nannou, M. E. Ur Rahman, and S. Shetty, "Enabling real-time AI in O-RAN: Deploying and measuring AI inside a near-RT

- RIC xApp,” in *Proc. 35th Int. Conf. on Computer Communications and Networks (ICCCN)*, Jul. 2026, pp. 1–9. doi: 10.1109/ICCCN69946.2026.11662852
- [22] B. Nugraha, M. Hajizadeh, T. Niehoff, A. V. Jnanashree, T. V. Phan, D. Triantafyllou, O. Krause, M. Mieth, K. Moessner, and T. Bauschert, “A comprehensive 5G dataset for control and data plane security and resource management,” in *Proc. IEEE International Conference on Cyber Security and Resilience (CSR)*, 2025, pp. 326–333. doi: 10.1109/CSR64739.2025.11130023
- [23] L. Bonati, M. Polese, S. D’Oro, S. Basagni, and T. Melodia, “OpenRAN Gym: AI/ML development, data collection, and testing for O-RAN on PAWR platforms,” *Computer Networks*, vol. 220, p. 109502, 2023. doi: 10.1016/j.comnet.2022.109502
- [24] L. Bonati, S. D’Oro, S. Basagni, and T. Melodia, “SCOPE: An open and softwareized prototyping platform for NextG systems,” in *Proc. ACM MobiSys*, 2021, pp. 415–426. doi: 10.1145/3458864.3466863
- [25] N. Nikaein, M. K. Marina, S. Manickam, A. Dawson, R. Knopp, and C. Bonnet, “OpenAirInterface: A flexible platform for 5G research,” *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 5, pp. 33–38, 2014. doi: 10.1145/2677046.2677053
- [26] I. Gomez-Miguel, A. Garcia-Saavedra, P. D. Sutton, P. Serrano, C. Cano, and D. J. Leith, “srsLTE: An open-source platform for LTE evolution and experimentation,” in *Proc. ACM WiNTECH*, 2016, pp. 25–32. doi: 10.1145/2980159.2980163
- [27] ETSI, “Publicly available specification (PAS); O-RAN security threat modeling and risk assessment (O-RAN.WG11.Threat-Modeling.O-R003-v03.00),” European Telecommunications Standards Institute, ETSI TR 104 106 V3.0.0, Jun. 2025. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/104100_104199/104106/03.00.00_60/tr_104106v030000p.pdf
- [28] M. Liyanage, A. Braeken, S. Shahabuddin, and P. Ranaweera, “Open RAN security: Challenges and opportunities,” *Journal of Network and Computer Applications*, vol. 214, p. 103621, 2023. doi: 10.1016/j.jnca.2023.103621
- [29] D. Mimran, R. Bitton, Y. Kfir, E. Klevansky, O. Brodt, H. Lehmann, Y. Elovici, and A. Shabtai, “Security of open radio access networks,” *Computers & Security*, vol. 122, p. 102890, 2022. doi: 10.1016/j.cose.2022.102890
- [30] J. Groen, S. D’Oro, U. Demir, L. Bonati, M. Polese, T. Melodia, and K. R. Chowdhury, “Implementing and evaluating security in O-RAN: Interfaces, intelligence, and platforms,” *IEEE Network*, vol. 39, no. 1, pp. 227–234, 2025. doi: 10.1109/MNET.2024.3434419
- [31] H. Fard, I. Komarov, and G. Wunder, “Cross-layer intrusion detection in 5G O-RAN: Gains and limits of fusing radio telemetry with network flow records,” arXiv:2606.22450, 2026. [Online]. Available: <https://arxiv.org/abs/2606.22450>
- [32] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, “Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset,” *Future Generation Computer Systems*, vol. 100, pp. 779–796, 2019. doi: 10.1016/j.future.2019.05.041
- [33] N. Moustafa, “A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets,” *Sustainable Cities and Society*, vol. 72, p. 102994, 2021. doi: 10.1016/j.scs.2021.102994
- [34] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, “Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning,” *IEEE Access*, vol. 10, pp. 40 281–40 306, 2022. doi: 10.1109/ACCESS.2022.3165809
- [35] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, “CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment,” *Sensors*, vol. 23, no. 13, p. 5941, 2023. doi: 10.3390/s23135941
- [36] I. Sharafaldin, A. Habibi Lashkari, and A. A. Ghorbani, “Toward generating a new intrusion detection dataset and intrusion traffic characterization,” in *Proc. 4th Int. Conf. on Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116. doi: 10.5220/0006639801080116
- [37] S. Samarakoon, Y. Siriwardhana, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Yliantila, “5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network,” *IEEE DataPort*, 2022. doi: 10.21227/xtep-hv36
- [38] Y. Siriwardhana, S. Samarakoon, P. Porambage, M. Liyanage, S.-Y. Chang, J. Kim, J. Kim, and M. Yliantila, “Descriptor: 5G wireless network intrusion detection dataset (5G-NIDD),” *IEEE Data Descriptions*, vol. 2, pp. 358–369, 2025. doi: 10.1109/IEEEDATA.2025.3592888
- [39] F. A. Zadeh, A. Civciss, V. Ravihansa, C. Sandeepa, and M. Liyanage, “Descriptor: 5G open radio access network multi-modal intrusion detection dataset (NetsLab-5GORAN-IDD),” *IEEE Data Descriptions*, vol. 2, pp. 348–357, 2025. doi: 10.1109/IEEEDATA.2025.3614167
- [40] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, “NetFlow datasets for machine learning-based network intrusion detection systems,” in *Big Data Technologies and Applications (BDTA/WiCON)*, LNICST, vol. 371, 2021, pp. 117–135. doi: 10.1007/978-3-030-72802-1_9
- [41] M. Sarhan, S. Layeghy, and M. Portmann, “Towards a standard feature set for network intrusion detection system datasets,” *Mobile Networks and Applications*, vol. 27, no. 1, pp. 357–370, 2022. doi: 10.1007/s11036-021-01843-0
- [42] G. Engelen, V. Rimmer, and W. Joosen, “Troubleshooting an intrusion detection dataset: the CICIDS2017 case study,” in *Proc. IEEE Security and Privacy Workshops (SPW)*, 2021, pp. 7–12. doi: 10.1109/SPW53761.2021.00009
- [43] R. Flood, G. Engelen, D. Aspinall, and L. Desmet, “Bad design smells in benchmark NIDS datasets,” in *Proc. IEEE European Symposium on Security and Privacy (EuroS&P)*, 2024, pp. 658–675. doi: 10.1109/EuroSP60621.2024.00042
- [44] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, p. 100804, 2023. doi: 10.1016/j.patter.2023.100804
- [45] G. Apruzzese, P. Laskov, E. Montes de Oca, W. Mallouli, L. Bdralo Rapa, A. V. Grammatopoulos, and F. Di Franco, “The role of machine learning in cybersecurity,” *Digital Threats: Research and Practice*, vol. 4, no. 1, pp. 1–38, 2023. doi: 10.1145/3545574
- [46] D. Kus, E. Wagner, J. Pennekamp, K. Wolsing, I. B. Fink, M. Dahlmans, K. Wehrle, and M. Henze, “A false sense of security? Revisiting the state of machine learning-based industrial intrusion detection,” in *Proc. ACM Cyber-Physical System Security Workshop (CPSS)*, 2022, pp. 73–84. doi: 10.1145/3494107.3522773
- [47] L. D’hooge, T. Wauters, B. Volckaert, and F. De Turck, “Inter-dataset generalization strength of supervised machine learning methods for intrusion detection,” *Journal of Information Security and Applications*, vol. 54, p. 102564, 2020. doi: 10.1016/j.jisa.2020.102564
- [48] M. Saerens, P. Latinne, and C. Decaestecker, “Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure,” *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002. doi: 10.1162/089976602753284446
- [49] Z. C. Lipton, Y.-X. Wang, and A. J. Smola, “Detecting and correcting for label shift with black box predictors,” in *Proc. 35th Int. Conf. on Machine Learning (ICML)*, 2018, pp. 3122–3130.
- [50] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [51] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, “Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift,” in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [52] T. Fawcett and A. Niculescu-Mizil, “PAV and the ROC convex hull,” *Machine Learning*, vol. 68, no. 1, pp. 97–106, 2007. doi: 10.1007/s10994-007-5011-0
- [53] S. Rabanser, S. Günnemann, and Z. C. Lipton, “Failing loudly: An empirical study of methods for detecting dataset shift,” in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [54] B. Sun, J. Feng, and K. Saenko, “Return of frustratingly easy domain adaptation,” in *Proc. 30th AAAI Conf. on Artificial Intelligence*, 2016, pp. 2058–2065. doi: 10.1609/aaai.v30i1.10306
- [55] C. Nadeau and Y. Bengio, “Inference for the generalization error,” *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003. doi: 10.1023/A:1024068626366
- [56] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. doi: 10.1023/A:1010933404324
- [57] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. doi: 10.1145/2939672.2939785
- [58] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://jmlr.org/papers/v12/pedregosa11a.html>
- [59] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. Conf. on*

- Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734. doi: 10.3115/v1/D14-1179
- [60] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. 8th IEEE Int. Conf. on Data Mining (ICDM)*, 2008, pp. 413–422. doi: 10.1109/ICDM.2008.17
- [61] X. Jin, L. E. Li, L. Vanbever, and J. Rexford, “SoftCell: Scalable and flexible cellular core network architecture,” in *Proc. 9th ACM Conf. on Emerging Networking Experiments and Technologies (CoNEXT)*, 2013, pp. 163–174. doi: 10.1145/2535372.2535377
- [62] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017, pp. 4765–4774. [Online]. Available: <https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>

ARSHAD ALI is with the Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia.

ADEEL AHMAD is with the Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia.

• • •